Dynamic Pricing Under Self-Exciting Arrival Processes

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We study dynamic pricing under a stochastic, non-linear, self-exciting demand arrival process over a finite sales horizon. We adopt such a correlated demand process to capture the phenomenon that customers who have made a purchase can inform and excite future customers to arrive. Specifically, the stochastic arrival intensity is stimulated with an immediate boost after any purchase and gradually decays over time between two consecutive purchases. The arrival intensity is also affected by the market phase the seller operates in, a growth or maturity stage, which convexly or concavely shapes how many new customers informed customers can stimulate. We show that the optimal pricing policy depends on the current time and excitement level, which measures the currently accumulated influence that previously informed customers exert and is shown to be the only sufficient statistic for our problem. Over time, the seller tends to initially set a low price to attract early customers who can excite later arrivals and later increase the price to maximize profits. Perhaps surprisingly, for a given time, in the growth stage of the market, the optimal price rises with the excitement level, and conversely, in the maturity stage of the market, the optimal price drops with the excitement level. Moreover, we show that adopting the optimal time-varying (open-loop) pricing policy solved from the deterministic problem (without resolving the problem) leads to only a constant performance loss from the optimal dynamic (i.e., contingent) pricing policy, which would not hold for the classic revenue management problem with a capacity constraint. Finally, we extend our analysis to account for general information dissemination, discrete prices, and multiple products in an assortment.

1. Introduction

In recent years, firms have increasingly leveraged customer-generated content to enhance their sales and pricing policies. For instance, customers' enthusiastic sharing of their purchases has become a crucial propeller of the sales process. The organic reach and authenticity of customer-shared content can enhance the brand's visibility and credibility, enticing new customers and driving sales. By monitoring social media platforms, firms can identify trends and adjust pricing policies in real-time

to maximize profits. This reactive approach has proven effective, as customers create and share a significant portion of product-related content on platforms like Instagram, Facebook, and TikTok.

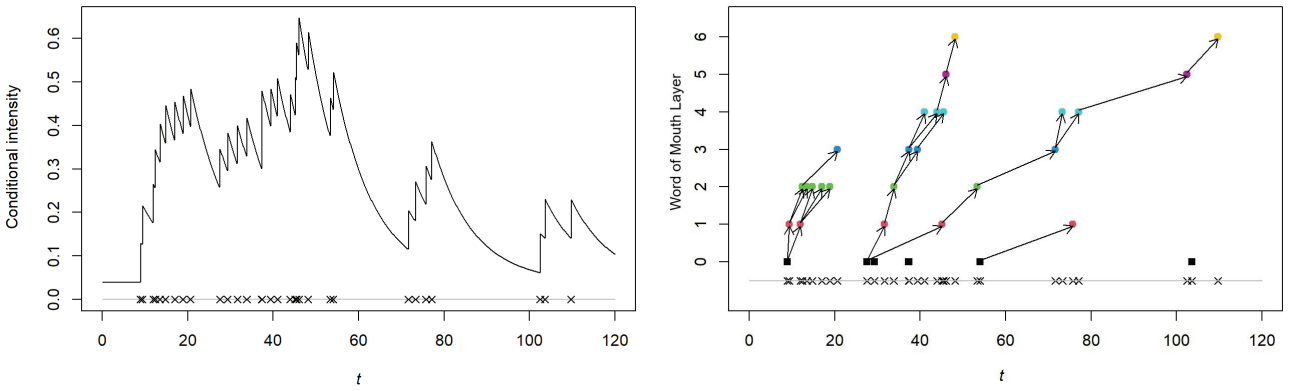
However, customer-shared content on social media platforms can experience a decline in relevance and engagement over time as new content emerges and older posts become less visible. The duration of user engagement, measured through metrics such as likes, comments, saves, and views, varies across different platforms. For instance, on Snapchat, engagement drops off almost immediately, exhibiting what can be termed "Instant Decay." On Twitter, the engagement span is approximately 43 minutes. Facebook content tends to engage users for about 76 minutes, while Instagram posts remain relevant for up to 19.75 hours. LinkedIn posts have a longer lifespan, with engagement persisting for about 24 hours (Graffius 2024).

The rapid growth of social media, with the network effects of word-of-mouth marketing and associated content decay, has significantly shaped how customers influence potential buyers. Customers who have purchased a product share their experiences within their social networks. This sharing leverages the social proof provided by satisfied customers to attract new buyers. However, as mentioned, this shared content can quickly decay and lose its impact. Companies need to respond actively to such a dynamic process of customer influence.

We observe how customers influence potential buyers using a real dataset. For example, we have a good model fit when calibrating the arrival process of a sunscreen product for March 8th on JD.com, one of China's largest retailers, by adopting a self-exciting process, also known as the Hawkes process. Each purchase by a customer stimulates additional purchases. Such a stimulation can gradually decay over time, as shown in Figure 1(a). In addition, we simulate a sample path of the customer-generating process for this product to illustrate the transition from one "generation" of purchased consumers to the next one being influenced, as shown in Figure 1(b). In this figure, cross signs represent the arrival of purchased customers, while arrows represent the transitions between generations.

When customer arrivals are self-exciting, the firm may want to adjust the product price as the dynamic purchase process evolves. It is challenging to balance the current period's profit with future potential profit in such a dynamic setting. Lower prices can attract more buyers and foster future sales, while higher prices can improve immediate profitability. It is essential to find a nuanced balance to optimize both immediate payoffs and future benefits.

In this paper, we study how a seller should dynamically adjust the selling price under a generalized self-exciting customer arrival process where the purchased customers would excite potential



(a) The arrival intensity of purchased customers

(b) The "generations" of customers under the word-of-mouth effect

Figure 1 Sales process on JD.com

future customers. Each new purchase continuously stimulates additional purchases, thereby increasing the stochastic arrival intensities, a feature we refer to as self-excitement of the arrival process. Moreover, such stimulation may decay over time because the influences of earlier purchases tend to become less significant as the associated social media posts become less fresh. Upon arrival, a customer decides whether to make a purchase based on the price at that moment. We formulate this problem as a dynamic program and study the structural properties of the optimal pricing policy. We show that the optimal pricing policy depends on the current time and excitement level, which measures the currently accumulated influence that previously informed customers exert and is shown to be the only sufficient statistic for our dynamic pricing problem. We fully characterize how the optimal pricing policy depends on the current time and excitement level.

We first demonstrate that the optimal pricing policy consistently encourages a higher purchase rate as the excitement level increases. This finding implies that at any time if the number of informed customers becomes higher, the optimal purchase rate should also be higher; although the optimal pricing can either increase or decrease in the excitement level, the optimal purchase rate always increases as the excitement level rises, even if the seller decides to increase the price.

Next, we show that the optimal price may not be monotone with respect to the excitement level. This non-monotonic behavior differs fundamentally from results in the existing revenue management (RM) literature, where the optimal price is typically constant without capacity constraints or is monotone in the capacity level. As the excitement level increases, it might be expected that the

seller would raise the price to maximize profit. However, we find the optimal price increases (resp., decreases) with respect to the excitement level depending on whether the market is in a growth (resp., maturity) phase. During a rapid growth phase of a product, the optimal price decreases with the excitement level because a small increase in the excitement level causes a significant expansion of the informed customer base, thus convexly increasing potential future arrivals. Therefore, when the excitement level is higher, the seller should reduce the price further to build an even larger pool of informed customers. Conversely, during a maturity phase of a product, the optimal price decreases with the excitement level, as the limited pool of potential customers diminishes the benefit of future arrivals. In this case, when the excitement level is higher, the seller increases the price to generate higher immediate profit and exploit the larger pool of existing informed customers. With linear self-exciting, we show the optimal intensity is independent of previous purchases, which can be simply obtained by solving two ordinary differential equations (ODEs).

Third, we demonstrate that the optimal price process exhibits an upward trend, technically known as a submartingale, when there is no excitement decay. At the beginning of the sales horizon, the seller tends to set a low price to attract purchases, thereby generating additional potential arrivals in the future. As the end of the sales horizon approaches, the seller tends to raise the price to make immediate profits. As prices are expected to rise stochastically over time, such an upward trend aligns with the practice of offering early-bird discounts each morning to attract a large customer base quickly in anticipation of influenced arrivals by social media posts picking up during the day. Moreover, the increasing trend may assist pricing managers in checking and evaluating the effectiveness of their existing pricing policy.

Computing the optimal pricing policy in our setting can be challenging. Unlike traditional RM problems where inventory is finite and decreases over time, the excitation level in our problem can be uncapped. This makes numerical computation challenging, as a large number of scenarios must be accounted for. This would be especially difficult for problems in which the base arrival rate is large. To search for easy-to-compute heuristics and study their performances, we consider a series of problems where we scale up the arrival rate by n. We show that the expectation of the excitement level grows at the rate of n if the self-exciting function is Lipschitz continuous. Under the latter innocuous assumption, the variance of the optimal intensity vanishes to 0 at a rate of $1/n$. Thus, as the scale of the problem increases, the optimal intensity converges to its expected value in probability. As the expectation of the optimal intensity is, in general, intractable, we are inspired to seek deterministic heuristics where the intensity over time is predetermined and does not change contingently on the excitement level.

In traditional RM problems, a commonly used deterministic heuristic is derived from the corresponding deterministic problem where customers arrive in a fluid fashion. We take a similar approach and study the deterministic version of our problem. For the traditional inventory capacitated RM problem, Wang and Wang (2022) show that the difference in revenues between the stochastic and deterministic problem is at the magnitude of $\ln(n)$. However, we prove that the difference in our setup is bounded by a constant independent of n. We then apply the deterministic heuristic, which is a time-varying (open-loop) pricing policy, to the stochastic problem and show that its performance loss is also a constant independent of n, whereas the performance loss of such a heuristic is at the scale of $\sqrt{n}$ for the traditional RM problem (see Gallego and Van Ryzin 1994).

We then extend our base model to three extensions: general information spreading, discrete prices, and multiple products. First, in the general information-spreading model, informed customers are influenced not only by prior purchases but also by non-purchasing customers who have arrived earlier. We show that the optimal price is non-monotone in the excitation intensity of leaving customers. Second, we establish that the synthetic discrete price heuristic is not asymptotically optimal, which is different from the RM literature. Specifically, any set of K discrete prices fails to synthesize and align with the optimal price, even if their expected price matches the optimal price. Therefore, pricing managers had better constantly monitor the market and set prices without constraining themselves within a set of predetermined prices. Lastly, for a customer making a selection among multiple products, we show that most of our findings from the base model remain applicable for this setting of multi-product pricing, in which one product's purchase can excite future arrivals who may end up purchasing another product.

Finally, we conduct an extensive numerical study to derive further insights. First, we investigate the necessity of considering self-excitement when the seller makes pricing decisions. Our findings indicate that neglecting self-excitation leads to significant profit loss for the seller. This insight underscores the need for pricing managers to consider the word-of-mouth effect in their pricing strategies. Second, we quantify the value of dynamic (i.e., contingent) pricing by analyzing the expected profit and comparing it against static pricing. Our results demonstrate that implementing optimal dynamic pricing can significantly enhance the seller's profit by changing prices based on information dissemination dynamics over social networks. Third, we study the performances of our theoretically examined deterministic heuristic, obtained from solving the deterministic counterpart of the original stochastic problem, and observe its superior performance in achieving profitability.

2. Literature Review

This paper is closely related to four distinct streams of literature, namely, self-exciting process, correlated customer arrivals in RM, pricing under network effect, and asymptotic analysis in RM. We will briefly review each of these areas and then summarize our contributions. First, our paper is related to the literature on the self-exciting process, also known as the Hawkes process, which was first introduced by Hawkes (1971). Hawkes introduces a family of models for stochastic point processes known as self-exciting and mutually exciting point processes. The process shows the critical property that the occurrence of any event increases the probability of the following events occurring. Brémaud and Massoulié (1996) extend this to a nonlinear self-exciting (Hawkes) process, proving the existence of a unique stationary version and the convergence of non-stationary processes. Additionally, Cui et al. (2020) propose an elementary method to solve the self-exciting process. Within the operations management literature, Daw and Pender (2018) study queues where customers arrive according to a Hawkes process. Daw et al. (2024) model the interaction between customers and agents using a bivariate marked Hawkes process. The authors also introduce a simple modified policy based on the Hawkes model, which has better performance compared to traditional customer routing policies. None of these studies have linked the self-exciting process with customer arrivals driven by a seller's pricing decisions. To our knowledge, ours is the first study to model customer arrivals as a self-exciting process in a pricing setting.

Second, our paper is related to the literature on correlated customer arrivals in RM. Typically, customers are assumed to arrive following a Poisson arrival process (Gallego and Van Ryzin 1994). In our setting, arrivals are also influenced by previous purchases. Other studies have modeled customer arrivals using a Markov process. Rustichini and Wolinsky (1995) investigate pricing decisions under a Markov demand process with constant transition probabilities. Aviv and Pazgal (2005) investigate dynamic pricing with a partially observed Markovian process, allowing the seller to update probabilities periodically. Keskin and Li (2024) study a firm's dynamic pricing in the context of unknown and time-varying customer preferences for quality. Jiang (2023) considers a network RM decision and assumes customer arrivals over different periods follow a time-inhomogeneous Markov chain. Our approach differs by considering a continuous time setting rather than making decisions over discrete time epochs. Specifically, we introduce an excitement level to capture the continuous influence of existing customers on new arrivals, which can be viewed as the state in a continuous-time Markov process. Additionally, we demonstrate the monotonicity properties of the optimal pricing policies and show that the price process tends to increase over time.

More recently, a third stream of work aims to address large demand volume and large variability. Bai et al. (2023) consider a demand model where customers arrive according to a Markov process, thus allowing the total demand to follow an arbitrary distribution. The authors propose a fluid approximation of the RM problem that becomes asymptotically tight as resource capacities increase. Li et al. (2023) study an RM model in which customers arrive in multiple stages in the selling horizon. They present asymptotically optimal policies based on fluid approximation when the number of stages and the resource capacities are scaled up. In contrast to those two papers assuming discrete time and capacity constraints, our work considers a self-exciting arrival process in continuous time and no capacity constraint. Moreover, in those papers, the demand for each stage depends on that of the previous stage. In our model with self-exciting properties, current demand is dependent on all previous purchases as well as the timing of those purchases. That is, unlike theirs, it is not enough to know and record the total number of all previous arrivals. Instead, in our setting, the optimal policy also needs to know the timing of all previous arrivals due to the influence decay effect of previous purchases. In our setting, we characterize the static and stochastic structural properties of the optimal pricing policy. Without a capacity constraint, we show the deterministic open-loop heuristic derived directly from the fluid counterpart of the stochastic problem (without any adjustment) leads to an expected profit only a constant away from the optimal profit in the stochastic problem.

Third, our paper contributes to the literature on pricing under network effects. Earlier studies in this field focused on static systems. For example, Dhebar and Oren (1986) characterize optimal pricing for new products considering network effects on consumption. Bloch and Quérou (2013) analyze a monopoly seller's optimal pricing under network effects among neighbors, demonstrating the relationship between price and consumer centrality. Wang and Wang (2017) analyze consumer choice models with network externalities, detailing choice probabilities, comparative statics, and assortment optimization. Nosrat et al. (2021) investigate pricing decisions influenced by experiential network effects, providing a heuristic for near-optimal pricing. Moreover, Du et al. (2016) explore pricing strategies for a product assortment under strong network effects, revealing that the optimal strategy involves setting a low price for one product and high prices for others. Moreover, there are papers studying dynamic pricing under network effects. Xie and Sirbu (1995) examine dynamic pricing and its profitability for incumbents and entrants under network externalities and competition, considering the general network benefits dependent on all customers' behaviors. Ajorlou et al. (2018) study a monopolist's dynamic pricing strategy with word-of-mouth information spread over an infinite horizon in a deterministic setting, demonstrating that optimal prices drop to

zero infinitely often with negligible marginal costs. Unlike these static or dynamic but deterministic models, our research examines contingent dynamic pricing under demand uncertainty.

Another related stream of research focuses on network effects under uncertainty. Lin et al. (2021) integrate diffusion effects for online content into the candidate generation and promotion optimization problem, demonstrating the consistency and efficiency of D-OLS estimators. Jiang et al. (2022) examine the dynamic competition between promotional efforts and pricing by retailers, finding that equilibrium prices increase over time while promotional efforts decrease, highlighting the need for retailers to balance this trade-off intertemporally. Yang and Zhang (2022) explore dynamic pricing and inventory management where customer reviews influence future purchasing behaviors and characterize the optimal policy. Huang et al. (2022) analyze the value of price discrimination based on customers' positions in social networks, showing significant value in small networks but little value in large random networks. Ahn et al. (2023) analyze how a company can use an influencer to influence its followers to make purchases, finding that the noisy signal of product quality can benefit the company. Our study differs by adopting the self-exciting process to capture how past purchases excite future customer arrivals. In addition, Shao et al. (2024) investigate a practice where a customer asks friends to "haggle" over the price the customer desires. Unlike their approach, we assume all self-excited customers may make a purchase rather than assisting a focal customer in obtaining a free product.

Finally, in search of heuristics and to analyze their asymptotic performances, we draw inspiration from relevant papers in traditional RM, with which our problem shares similarities. Gallego and Van Ryzin (1994) adopt the optimal solution of the deterministic fluid model as the deterministic heuristic for the stochastic problem. They show that the deterministic heuristic is asymptotically optimal as the demand rate and the inventory are scaled up. As the scale of the problem n grows, the deterministic heuristic has a revenue loss at the magnitude of $\sqrt{n}$. To further reduce the gap, a commonly used heuristic is to periodically resolve the deterministic problem whose solution is then adopted in the stochastic problem (e.g. see Cooper 2002, Maglaras and Meissner 2006, Reiman and Wang 2008, Jasin and Kumar 2013). In particular, for the price-based RM problem, Jasin (2014) show that the frequent resolving heuristic reduces the revenue gap to the magnitude of $\ln(n)$. They also propose a linear rate correction heuristic that adjusts the prices with a predetermined formula and show that it achieves the same performance loss of $\ln(n)$ as the resolving heuristic. Chen et al. (2016) show that a heuristic that adjusts prices of only a small set of products can also achieve the same bound. Studying the dynamic pricing problem under a cascade click demand model where customers are presented with products sequentially, Najafi et al. (2024) show that

the deterministic heuristic is asymptotically optimal and has a performance loss at the magnitude of $\sqrt{n}$. They propose a cascade correction heuristic that updates the prices periodically and show that it achieves an expected performance loss at the magnitude of $\ln(n)$. Zhang et al. (2022) model the adoption of a new product as a continuous time Markov chain and study the dynamic pricing problem where the parameters of the adoption rate are unknown. They propose pricing algorithms that update the parameter estimation over time and show that the algorithms achieve a revenue loss at the magnitude of the logarithm of the market size.

3. Formulation and Preliminary Results

3.1. The Model

Consider a seller who dynamically adjusts prices to promote and sell a single product over a finite horizon, denoted as $[0, T]$, say 24 hours. There are two types of potential customers. The first type of customer, called organic customers, arrives randomly according to their own needs with an arrival rate φ. The second type of customer is excited by the previous customers who have made a purchase and had a strong incentive to inform others, e.g., by posting their purchased item on social media. Let $\{t_1, t_2, \ldots, t_k\}$ represent the realized and observed sequence of arrival times of purchasing customers up to time t. For a customer who arrives and makes a purchase at time t_i, she informs additional customers, called informed customers, who may arrive at a later time $t \geq t_i$, at a stochastic rate of $\alpha e^{-\beta(t-t_i)}$, where the excitation intensity α represents the baseline contribution to the arrival intensity of informed customers. As older posts become less visible, the decay parameter β captures the rate of time decay of the excitation of informed customers.[1] Thus, the aggregated stochastic intensity of informed customers is given by $\sum_{t_i<t} \alpha e^{-\beta(t-t_i)}$.[2]

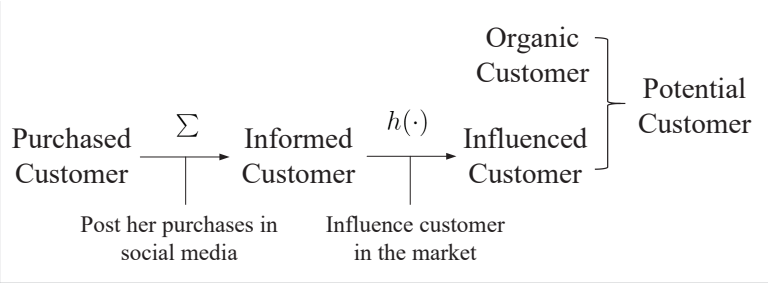


Figure 2 **General self-exciting process**

[1] Caro and Martínez-de Albéniz (2020) similarly model the decay in the attractiveness of content released over time.

[2] This aggregated intensity is originally introduced by Hawkes (1971). They use the exponential decay rate to describe the self-exciting process.

Informed customers can also influence other customers in the market, referred to as influenced customers. These influenced customers arrive with a stochastic intensity of $h(\sum_{t_i<t} \alpha e^{-\beta(t-t_i)})$. The function $h(\cdot)$ is strictly increasing, as a rise in the number of informed customers results in a higher arrival rate of influenced customers. Later in this section, we will demonstrate that the optimal pricing solutions exhibit different structural properties depending on the shape of $h(\cdot)$.

We now define the stochastic intensity of potential customers, which follows a generalized self-exciting process (see, e.g., Brémaud and Massoulié 1996 and Zhu 2015), as follows:

Definition 1 (General Self-Exciting Process). Let $t_1, t_2, \ldots, t_k \in [0, t)$ be the purchase time epochs up to time t. Potential customers arrive according to a point process N_t with the following stochastic intensity:

$$\Phi_t(\mathcal{H}_t) = \underbrace{\varphi}_{\text{organic}} + \underbrace{h(\overbrace{\sum_{t_i<t} \alpha e^{-\beta(t-t_i)}}^{\text{informed}})}_{\text{influenced}},$$

where the self-exciting function $h(\cdot)$ is strictly increasing.

The above definition of the potential demand process covers a few notable special cases. For instance, when $\alpha = 0$, the demand process reduces to a standard Poisson process. When β goes to infinity, the incentive to inform others quickly decays, and the demand process also reduces to a standard Poisson process. When $h(\cdot) = 0$, there is no influence made by the informed customers, and the demand process is again reduced to a standard Poisson process. When $h(x) = x$, informed customers themselves are potential customers without bringing in additional potential customers.

The seller influences market demand by dynamically adjusting its prices, p_t, over the time interval $t \in [0, T]$. When a potential customer arrives, she decides whether to make a purchase based on the current price. Given a price of p, the potential customer decides to buy the product with probability $\lambda(p)$. For each unit of sales, the seller incurs a marginal cost of c. Thus, the purchase rate of the product can be expressed as follows:

$$\Lambda_t(p) := \lambda(p) \left(\varphi + h \left(\sum_{t_i<t} \alpha e^{-\beta(t-t_i)} \right) \right). \tag{1}$$

Following the convention in RM (e.g., Gallego and Van Ryzin 1994), we use the purchase probability (also known as the demand rate) λ as the decision variable and make the following standard assumptions:

Assumption (R) (Regular Assumption).

(i) $\lambda(p) \geq 0$ *is differentiable and invertible in* p*;*

(ii) The profit rate $r(\lambda) = \lambda(p(\lambda) - c)$ is differentiable, bounded and strictly concave in λ.

Denote $[\underline{\lambda}, \bar{\lambda}]$ as the range of feasible purchase probabilities. The seller selects a pricing policy $u = \{\lambda_t \in [\underline{\lambda}, \bar{\lambda}] : t \in [0, T]\}$ that depends on the realized purchases $\{N_s\}_{s \le t}$ up to time t, to maximize the expected profit from t onward

$$V_t(\{N_s\}_{s \le t}) := \max_u \mathbb{E}_u \int_t^T (p(\lambda_s) - c) dN_s, \tag{2}$$

where N_t is a general Hawkes process whose rate is given by Equation (1).

3.2. Structure of Optimal Policy

Before presenting the structure of the optimal policy, we define the excitement level

$$x_t := \sum_{t_i < t} \alpha e^{-\beta(t - t_i)}.$$

Here, x_t is the cumulative influence intensity of informed customers, which is stimulated by an immediate increase after a purchase and gradually decays over time between two consecutive purchases. In the following lemma, we characterize the dynamics of x_t with a stochastic differential equation:

LEMMA 1. *x_t is characterized by the following differential equation with stochastic jumps:*

$$\frac{dx_t}{dt} = -\beta x_t dt + \alpha dN_t,$$

where N_t is a point process with intensity $\Lambda_t(p) = \lambda_t(\varphi + h(x_t))$, and $x_0 = 0$.

Lemma 1 characterizes the excitement level x_t, which increases by α after every purchase and decays over time at the rate of β between two consecutive purchases. Next, we characterize the total expected profit, which depends on the intensity of potential customers. Different from the constant arrival intensity in the traditional RM problem, the potential customer intensity in our setting is nonstationary. This intensity is a function of the timing of all previous purchases, $\{t_i : i < N_t\}$. Tracking the timing of each purchase would significantly increase the dimensionality. By defining the excitement level, we can simplify potential customers' intensity and present the Hamilton-Jacobi-Bellman(HJB) condition in the following proposition.

PROPOSITION 1 (HJB EQUATION). *$V_t(x) = V_t(\{N_s\}_{s \le t})$ and is determined by the following differential equation:*

$$-\frac{\partial}{\partial t} V_t(x) + x\beta \frac{\partial}{\partial x} V_t(x) = (h(x) + \varphi) \max_\lambda \{r(\lambda) + \lambda (V_t(x + \alpha) - V_t(x))\}, \tag{3}$$

with boundary condition $V_T(x) = 0$ for any $x \ge 0$.

In the right-hand side of the above equation, $V_t(x+\alpha)-V_t(x)$ can be viewed as the future impact of one additional purchase. The optimal intensity is determined to balance the current period profit and the future profit lift. We denote $\lambda_t^*(x)$ as the unique solution to Equation (3) and examine the structural properties of the optimal policy under Assumption (R) that the profit rate function is strictly concave. While Equation (3) in general does not have close-form solutions, we illustrate the optimal solution through the following special case.

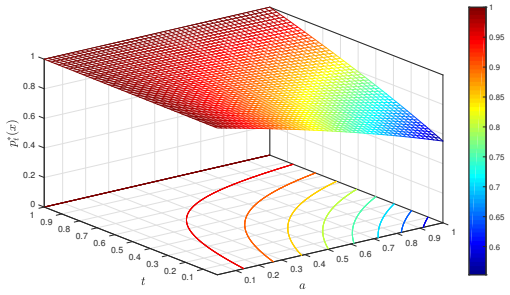


Figure 3 **Optimal price $p_t^*(x)$ over $t\in[0,1]$, $h(x)=x$, $c=0$, $\lambda(p)=ae^{-p}$, $\alpha=1$, and $\beta=0$**

Example 1. When $\beta=0$, $c=0$, $h(x)=x$, and the demand function $\lambda(p)=ae^{-p}$ where $a>0$,

$$\lambda_t^*(x)=\frac{a}{\alpha a(t-T)+e},\qquad p_t^*(x)=\ln(\alpha a(t-T)+e).$$

The optimal expected profit is given by:

$$V_t(x)=\frac{1}{\alpha}(1-\ln(\alpha a(t-T)+e))x-\frac{\varphi}{\alpha}(\ln(\alpha a(t-T)+e)).$$

In this special case, the optimal purchase probability and price are independent of the excitement level. This finding is intriguing and somewhat counterintuitive. It suggests that, regardless of the number of previous customers, the optimal purchase probability and price only depend on the time in the horizon. □

Next, we consider the pricing policy without self-excitement. When $\alpha=0$ or $\beta\to\infty$, the customer arrival process is reduced to a time-homogeneous Poisson process. We have the following result.

Lemma 2 (No Self-Excitement Benchmark). *If $\alpha=0$ or $\beta\to\infty$, customers arrive following a Poisson process, the optimal intensity is constant, i.e., $\lambda_t^*=\lambda^0=\arg\max_{\lambda\in[\underline{\lambda},\bar{\lambda}]}\{r(\lambda)\}$.*

Without self-excitement, the optimal intensity (price) remains constant and is set to maximize the profit rate $r(\lambda)$, which is the one for Poisson arrivals. However, with self-excitement, as mentioned, the term $V_t(x+\alpha)-V_t(x)$ on the right-hand side of Equation (3) represents the future

impact of one additional purchase. With this future effect considered, the optimal policy generally does not myopically maximize the immediate profit rate $r(\lambda)$. Instead, it balances immediate profit with the future benefit of an increased excitement level. Next, we present monotonicity properties of the expected profit $V_t(x)$.

Proposition 2 (Monotonicity Properties of Value Function). *(i) $V_t(x)$ is decreasing in t and strictly increasing in x; (ii) $V_t(x)$ is strictly increasing in α and strictly decreasing in β.*

With everything else equal, from Proposition 2(i), the expected profit-to-go decreases as the time elapses and gets closer to the end of the sales horizon. Moreover, the expected profit-to-go increases with the current excitement level being higher. Proposition 2(ii) demonstrates that an increase in the excitation intensity α attracts additional customers, thereby increasing the expected profit. Similarly, a decrease in the decay rate β leads to more customer arrivals and, consequently, increases the expected profit. A direct implication of these monotonicity properties is that the optimal expected profit under the self-exciting process is larger than that under the Poisson arrivals (without self-excitement). Additionally, the intensity of the self-exciting arrivals is larger than that of the Poisson process. We now proceed to examine the properties of the optimal purchase rate.

Proposition 3. *The optimal purchase rate $\Lambda_t^*(x) := \lambda_t^*(x)(\varphi + h(x))$ increases in x.*

Proposition 3 states that, under the seller's optimal pricing policy, the purchase rate increases with the excitement level. This optimal policy consistently encourages additional purchases as the number of informed customers grows. Notably, this proposition holds without any specific assumptions about the self-exciting function $h(x)$. Later, in Theorem 2, we find that the optimal purchase probability $\lambda_t^*(x)$ and the optimal price can either increase or decrease in the excitement level depending on the shape of $h(\cdot)$. However, we find that the optimal purchase rate always increases in the excitement level, even if the seller decides to increase the prices for a higher excitement level. The following theorem demonstrates the monotonicity of the optimal solutions with respect to the elapsed time t.

Theorem 1 (Static Monotonicity in t). *$\lambda_t^*(x)$ (resp., $p_t^*(x)$) and $\Lambda_t^*(x)$ decrease (resp., increases) in t for any $x \geq 0$.*

Theorem 1 indicates that the optimal purchase probability declines in the elapsed time, which implies that the optimal price correspondingly increases in the elapsed time. For the same excitement level, at the beginning of the sales horizon, the seller sets a lower price to attract more customers, thereby generating additional potential arrivals in the future, while as the end of the

sales horizon approaches, the seller raises the price to gain immediate profits. The driving force behind this is that the same exciting level has a higher value in driving future sales at the earlier horizon than the later.

In Proposition 3, we demonstrate that the optimal purchase rate increases with the excitement level x. The purchase rate is determined by the purchase probability, $\lambda_t^*(x)$, which is determined by the seller, and the number of informed customers, $\varphi + h(x)$. Since the number of informed customers increases with the excitement level x, the monotonicity of the optimal purchase probability in the excitement level is not immediately evident from Proposition 3. To address this, we establish the monotonicity of the optimal purchase probability $\lambda_t^*(x)$ in the following theorem and explore the convexity and concavity of the expected profit $V_t(x)$ with respect to the excitement level x.

THEOREM 2 (STATIC MONOTONICITY IN x). *For any $t \in [0, T)$:*

(i) If $h(x)$ is strictly convex, $V_t(x)$ is strictly convex in x, and $\lambda_t^*(x)$ (resp., $p_t^*(x)$) strictly increases (resp., decreases) in x.

(ii) If $h(x)$ is strictly concave, $V_t(x)$ is strictly concave in x, and $\lambda_t^*(x)$ (resp., $p_t^*(x)$) strictly decreases (resp., increases) in x.

(iii) If $h(x)$ is linear, $V_t(x)$ is linear in x, and $\lambda_t^*(x)$ (hence, $p_t^*(x)$) is independent of x.

The optimal policy balances maximizing immediate profit and the pool of potential customers for future benefits. On the one hand, when the seller lowers the purchase probability (equivalently, increases the price), the total immediate purchase rate decreases. This results in a gradual reduction in the excitement level x and, consequently, a decrease in the number of informed customers, which lowers future profits. On the other hand, increasing the price also leads to an immediate increase in the profit rate. Theorem 2 demonstrates that the optimal purchase probability increases (resp., decreases) with respect to the excitement level x if and only if the self-exciting function is convex (resp., concave). As the excitement level increases, one might expect the seller to raise the price to maximize profit. However, we find that the optimal price may actually decrease with a higher excitement level. A convex self-exciting function $h(x)$ can correspond to a stage of rapid market growth for a product. In this phase, a small increase in the excitement level x results in a significant expansion of the informed customer base, thereby increasing potential future arrivals. These future arrivals result in substantial future benefits. Therefore, when the excitement level x is higher, the seller reduces the price further to build a larger pool of informed customers.

Conversely, a concave self-exciting function $h(x)$ can reflect a phase of market maturity for a product. In this phase, there is a limited pool of potential customers, and the benefit of increasing

the excitement level x diminishes. In this case, the optimal purchase probability (resp., price) decreases (resp., increases) with the excitement level, leading to an immediate increase in the profit rate while yielding limited future benefits. Thus, when the excitement level x is higher, the seller increases the price to generate higher immediate profit and exploit the higher excitement level.

Pricing managers should regularly analyze market conditions to determine whether the market is in a growth or maturity phase. In a growth phase, engaging early adopters who are likely to spread positive word-of-mouth is crucial; using incentives, discounts, or loyalty programs can encourage sharing experiences and attract more customers. Conversely, in a maturity phase, when the existing customer base is larger, the seller can exploit it by charging higher prices. By following these strategies, pricing managers can effectively leverage the dynamic interplay between pricing and customer engagement to balance both immediate and future profits optimally.

Directly from Theorem 2, we see that when the self-exciting function $h(x)$ is linear, the value function is also linear in x, and the optimal purchase probability is independent of x. We further investigate this case in Corollary 1.

Corollary 1 (Linear Self-Exciting Function). *If $h(x) = h_1 x + h_0$ for $h_1 \geq 0$ and $h_0 \geq 0$, the value function $V_t(x) = \mu_t x + \omega_t$, where μ_t and ω_t satisfy the following equations:*

$$\begin{aligned} \frac{d\mu_t}{dt} &= \beta\mu_t - h_1 \max_{\lambda}\{r(\lambda) + \lambda\alpha\mu_t\}, \\ \frac{d\mu_t}{dt} &= \beta\mu_t + \frac{h_1}{h_0 + \varphi}\frac{d\omega_t}{dt}, \end{aligned} \tag{4}$$

with $\mu_T = 0$ and $\omega_T = 0$. The optimal purchase probability $\lambda_t^* = \arg\max_{0\leq\lambda\leq\lambda(0)}\{r(\lambda) + \lambda\alpha\mu_t\}$, which is independent of x.

With linear self-excitement, an increase in the excitement level results in a proportional expansion of the informed customer base, leading to a corresponding increase in potential future arrivals. The change in the size of the informed customer pool is independent of the initial excitement level. Consequently, the seller's optimal pricing decision, which balances immediate profit with future benefits, also remains independent of the excitement level. This implies that pricing managers can implement a relatively stable pricing policy that fluctuates over time rather than contingent adaptions based on the excitement level. Such an open-loop pricing policy not only reduces the complexity of pricing decisions but also allows managers to not focus on the number of users making purchases and instead on consistent marketing activities over the sales horizon.

3.3. Optimal Price Trend

From Theorems 1 and 2, we characterize the monotonicity properties of the optimal price with respect to time t and the excitement level x. In the following theorem, we investigate the (stochastic) price trend over time by studying the average sample path of the optimal price. Denote by $g(p) = -\lambda'(p)/\lambda(p)$ the hazard rate of the customer valuation distribution and by $1/g(p)$ the Mills ratio.

THEOREM 3 (OPTIMAL PRICE TREND). *Assume that $1/g(p)$ is convex in p and $\beta = 0$. $\mathbb{E}[p_s^*(x_s)|\mathcal{F}_t] \geq p_t^*(x_t)$ for any $0 \leq t < s \leq T$.*

The condition that $1/g(p)$ is convex in p is satisfied by many common distributions such as normal, uniform, logistic, Weibull, and gamma with shape parameters no less than one (see Xu and Hopp 2009 and Pang et al. 2015). Theorem 3 shows that under the special case where the excitement level x does not decay over time, the optimal price process has an upstream trend.

The pattern of increasing prices over time aligns with the common practice of managers offering early-bird discounts at the beginning of the sales horizon and using phrases like "limited time offer" or "today only" to attract a larger customer base rapidly. As the sales period progresses and early sales simulate later arrivals through social effects, gradually increasing prices can take advantage of the larger customer base. Furthermore, this increasing price trend offers managers a guideline for evaluating their pricing policy. If prices gradually decrease throughout the day, it suggests that the implemented pricing policy may not be profitable.

4. Asymptotic Analysis

We have shown the structural properties of the optimal policy in the previous section. Since there is no general closed-form solution for the optimal intensity as a function of the states, a common approach to numerically calculate the optimal solution is to discretize t with a small step size $\delta > 0$ and compute $V_t(x)$ and $\lambda_t(x)$ backward from the end of the horizon T. However, such a calculation would require the value of $x + \alpha$ at $t + \delta$. Thus if we would like to find the optimal solution $\lambda_t(x)$ for any $t \geq 0$ and $x \leq \bar{x}$, we must iteratively calculate all of the value functions $\{V_t(x) : x \leq \bar{x} + \lceil \alpha t/\delta \rceil\}$. Significant computation effort may be needed to calculate the optimal intensity, especially for problems on a large scale.

We generalize the formulation of the original problem to study the heuristics and their performances when the scale of the problem is large. In particular, we consider a series of problems: in the n-th problem, the Poisson process N_t admits the intensity $\Lambda_t^{(n)} = n\lambda_t[\varphi + h(x_{t-}/n)]$. The system state x_t is given by $dx_t = -\beta x_t dt + \alpha dN_t$ and $x_0 = 0$. The terminal payoff at T is $V_T^{(n)}(x) = 0$ for any x. The problem we study in the previous sections corresponds to the problem where $n = 1$. We first provide asymptotic bounds of the expectation and variance of x_t with respect to n.

4.1. Asymptotic Bounds of $\mathbb{E}[x_t]$ and $\mathrm{Var}[x_t]$

As we study the problems where n is large, we would like to ensure that the problems are "properly" scaled: One would expect $\mathbb{E}[x_t] = O(n)$ as we scale Λ_t by n. However, this seemingly obvious property may not hold for general $h(\cdot)$ because of the self-exciting property. In particular, since x_t is unbounded, it is possible to have a small subset of sample paths where x_t/n is very large. For certain convex $h(x)$, it is possible that those small number of cases drive $\mathbb{E}[x_t]/n \to \infty$. In the following proposition, we provide a sufficient condition that ensures $\mathbb{E}[x_t]$ is bounded by n.

PROPOSITION 4. *If $h(x)$ is Lipschitz continuous, $\mathbb{E}[x_t^u] = O(n)$ for any Markov policy u.*

The Lipschitz condition in Proposition 4 ensures that the arrival rate $\Lambda_t^{(n)}$ does not increase "too fast." It is worth noting that the Lipschitz condition is also shown as a sufficient condition for the stability of the nonlinear Hawkes process in Brémaud and Massoulié (1996). It guarantees that the average intensity is finite over any time interval. Therefore, in the asymptotic analysis, we make the following additional assumption about $h(x)$:

ASSUMPTION (LD). $h(\cdot)$ is Lipschitz continuous and infinitely differentiable.

From Theorem 2, the optimal intensity λ_t^* is dependent on the excitement level x at time t. It is, therefore, a random variable that fluctuates depending on the demand realization. In turn, this would potentially lead to greater variance of the excitement level x_t^*. In the next proposition, we established the bounds of the variance of λ_t^* and x_t^*.

PROPOSITION 5. *Under the optimal policy, $\mathrm{Var}[\lambda_t^*] = O(1/n)$ and $\mathrm{Var}[x_t^*] = O(n)$.*

Proposition 5 demonstrates that the variance $\mathrm{Var}[\lambda_t^*]$ of the optimal intensity is "small" as the scale of the problem increases. Moreover, from Daw and Pender (2018), the variance of the Hawkes process is higher than that of the standard Poisson process due to the self-exciting property. Here, we show that its variance under a stochastic intensity is still the same magnitude of n as long as the variance of the intensity is not too large, which is true under the optimal policy, and thus, $\mathrm{Var}[x_t^*] = O(n)$.

The formulation of our problem shares similarities with traditional RM problems. However, the result in Proposition 5 may not hold in traditional RM problems, where sales are capacitated by a given inventory level. This is because even though the optimal sales rate λ_t^* would change slowly with respect to the inventory, when inventory drops to 0, λ_t^* must also drop to 0. As time t gets closer to the end of the selling horizon, the probability of sellout increases, causing an increase in the variance of λ_t^*.

Proposition 5 may not hold for a self-exciting function $h(x)$ that is not Lipschitz continuous. For a general function that is infinitely differentiable, we can construct a corresponding Lipschitz continuous function $\hat{h}(x)$ that approximates $h(x)$ by a linear extrapolation, i.e., with the tangent line when x is sufficiently large:

$$\hat{h}(x) = \begin{cases} h(x) & \text{if } x \le \bar{x}, \\ h(\bar{x}) + h'(\bar{x})(x - \bar{x}) & \text{if } x > \bar{x}. \end{cases}$$

The constructed function $\hat{h}(x)$ has a bounded first derivative, so it is Lipschitz continuous. By Proposition 5, if we choose $\bar{x} \ge 1 + \mathbb{E}[x_t^*]/n$, $\mathbb{P}(x_t^*/n \ge \bar{x}) \le \mathbb{P}(x_t^* \ge n + \mathbb{E}[x_t^*]) \le \mathrm{Var}[x_t^*]/n^2 = O(1/n)$ where the inequality is by Chebyshev's inequality. Such an approximation of $h(\cdot)$ only affects a very small set of cases. Thus, our Lipschitz continuity assumption is not overly restrictive.

4.2. Deterministic Heuristic and Profit Bounds

In a traditional stochastic RM problem, a popular heuristic is adopted from the deterministic version of the problem in which customers arrive in a fluid fashion. Gallego and Van Ryzin (1994) has shown that the deterministic heuristic is asymptotically optimal when the scale of the problem $n \to \infty$. The authors show that the revenue loss of adopting the deterministic heuristic is at most in the scale of $\sqrt{n}$ compared to the revenue of the deterministic problem as a benchmark. From our discussion in the previous section, the variance of the optimal sales rate λ_t^* increases as it gets closer to the end of the selling horizon. However, the deterministic heuristic has a fixed sales rate, so it cannot make an adjustment based on the random realization of demand. To improve the revenue gap, reoptimization heuristics are proposed to resolve the deterministic problem and update the sales rate at distinct times during the sales horizon. Jasin (2014) shows that the heuristic that resolves the deterministic problem at discrete time points $t = 1, 2, \ldots, n$ achieves a $\log(n)$ performance gap when compared to the revenue of the deterministic problem. Moreover, the author derives an alternative heuristic that resolves more frequently when it gets closer to the end and shows that such a heuristic achieves the same $\log(n)$ performance bound.

Unlike the traditional RM problems, the state variable x_t is uncapped in our problem. It can potentially increase quickly because of the self-exciting property. Nonetheless, in search for a well-performing heuristic, we also take a similar approach and investigate the deterministic problem where customers arrive in a fluid fashion with rate $\Lambda_t = n\lambda_t(\varphi + h(x_t/n))$. Here, x_t, unlike in the stochastic problem, is calculated deterministically for a given λ_t and x_t:

$$x_{t+\delta} = (1 - \Lambda_t\delta)(x_t - \beta x_t\delta) + \Lambda_t\delta(x_t + \alpha - \beta x_t\delta) = x_t + (\alpha\Lambda_t - \beta x_t)\delta = x_t + [\alpha n\lambda_t(\varphi + h(x_t/n)) - \beta x_t]\delta.$$

Rearranging the terms and taking $\delta \to 0$, we have $dx_t/dt = \alpha n \lambda_t(\varphi + h(x_t/n)) - \beta x_t$. Therefore, the deterministic version of our problem can be formulated as the following optimal control problem:

$$V^d(n) = \max_{\lambda_t} \ \Pi^d(\lambda_t) = \int_0^T n\lambda_t \left[\varphi + h\left(\frac{x_t}{n}\right)\right] \cdot (p(\lambda_t) - c)dt$$
$$\text{s.t.} \ \ \frac{dx_t}{dt} = \alpha n \lambda_t \left[\varphi + h\left(\frac{x_t}{n}\right)\right] - \beta x_t, \quad x_0 = 0.$$

In a traditional RM problem, the optimal deterministic arrival rate is fixed throughout the sales horizon. However, in our problem, the optimal rate varies in time t. Nevertheless, there exist numerical techniques (e.g., forward-backward sweep method) that allow us to calculate numerical solutions efficiently.

We now study the performance of the deterministic heuristic policy DH, where we adopt the optimal solution in the deterministic problem λ_t^d as the deterministic sales rate in the stochastic problem. From Proposition 5, we observe that the optimal λ_t^* has a "small" variance as the scale n increases. Based on this observation, we hypothesize that the profit difference introduced by using a non-adaptive intensity would also be small. In the following theorem, we bound the optimal expected profit $V^*(n)$ and the expected profit of the deterministic heuristic $V^{DH}(n)$ in relationship to the optimal profit $V^d(n)$ in the deterministic problem.

THEOREM 4 (PERFORMANCE GAP). *There exist two constants K_1 and K_2 independent of n, such that $|V^{DH}(n) - V^d(n)| \le K_1$ and $|V^*(n) - V^d(n)| \le K_2$.*

We provide an intuitive proof sketch for the above theorem: for any Markov policy u, $\{\mathbb{E}[\lambda_t^u]\}$ is a feasible solution in the deterministic problem. We consider a subclass of policies in which $\mathrm{Var}[\lambda_t^u] = O(1/n)$. We prove that for those policies, the difference between the expected profit $V^u(n)$ in the stochastic problem and the profit of using $\{\mathbb{E}[\lambda_t^u]\}$ in the deterministic problem is bounded by a constant, i.e., $V^u(n) = \Pi^d(\{\mathbb{E}[\lambda_t^u]\}) + O(1)$. Since the variance of the optimal policy $\mathrm{Var}[\lambda_t^*] = O(1/n)$ from Proposition 5, $V^*(n) = \Pi^d(\{\mathbb{E}[\lambda_t^u]\}) + O(1) \le V^d(n) + O(1)$. On the other hand, consider the deterministic heuristic DH: $V^*(n) \ge V^{DH}(n) = \Pi^d(\{\mathbb{E}[\lambda_t^{DH}]\}) + O(1) = V^d(n) + O(1)$.

The key assumption in the above arguments is that $\mathrm{Var}[\lambda_t^u] = O(1/n)$ for any t. We note that the static heuristic in the traditional RM problem with an inventory constraint does not satisfy such a condition: even though the sales rate is fixed under the static heuristic, the probability of sellout increases as it gets closer to the end of the sales horizon, which means that $\mathrm{Var}[\lambda_t^{DH}] = \Theta(1)$ as $t \to T$.

For the classic RM problem, Gallego and Van Ryzin (1994) show that $|V^{DH}(n) - V^d(n)| = O(\sqrt{n})$. Wang and Wang (2022) prove that $|V^*(n) - V^d(n)| = \Theta(\ln(n))$. However, in our problem, we demonstrate in Theorem 4 that both $|V^*(n) - V^d(n)|$ and $|V^{DH}(n) - V^d(n)|$ are bounded by a constant. As a direct implication, the performance loss of the deterministic heuristic is also bounded by a constant, i.e., $V^{DH}(n) - V^*(n) = O(1)$. For problems of large scales, the deterministic heuristic is not only asymptotically optimal but also has negligible performance loss. Since the performance loss of deterministic heuristics is bounded by a constant and becomes negligible for large-scale problems, pricing managers can confidently use these heuristics for large-scale decision-making without concern for efficiency loss. The superior performance of deterministic heuristics will be numerically tested and confirmed in Section 6.

5. Model Extensions

In this section, we extend our base model to account for general information spreading, discrete pricing, and multiple products. We explain how the earlier results can be applied to these new contexts.

5.1. General Information Spreading

In this subsection, we extend our base model to a general information-spreading setting. Suppose that, beyond being influenced by previous purchases, informed customers are also informed by non-purchasing customers. Denote by $\{t_1^O, t_2^O, \ldots, t_i^O\}$ the observed arrival times of non-purchasing customers up to time t. When a customer arrives at time t_i^O and turns out not to make a purchase, she subsequently informs additional customers at a rate of $\alpha^O e^{-\beta(t-t_i^O)}$ to arrive at a later time $t \geq t_i^O$. Motivated by that non-purchased customers tend not to have a strong incentive to inform others than a purchased customer, we assume that the excitation intensity from non-purchasing customers is no more than that from purchasing customers, i.e., $\alpha^O \leq \alpha$ (Ajorlou et al. 2018 and Banerjee et al. 2013). Then, the intensity of potential customers can be expressed as follows:

$$\Phi_t(\mathcal{H}_t) = \varphi + h\left(\sum_{t_j<t} \alpha e^{-\beta(t-t_j)} + \sum_{t_i^O<t} \alpha^O e^{-\beta(t-t_i^O)}\right).$$

As discussed in the base model, $h(\cdot)$ is assumed to be positive and strictly increasing. If the price is set to p, each potential customer decides to buy with a non-negative bounded probability $\lambda(p)$. Therefore, the arrival rate of customers who decide to buy is,

$$\Lambda_t^b(p) = \lambda(p)(\varphi + h(\sum_{t_j<t} \alpha e^{-\beta(t-t_j)} + \sum_{t_i^O<t} \alpha^O e^{-\beta(t-t_i^O)})).$$

Each potential customer decides to leave with a probability of $1-\lambda(p)$. Consequently, the arrival rate of customers who choose not to purchase is,

$$\Lambda_t^O(p) = (1-\lambda(p))(\varphi + h(\sum_{t_j<t} \alpha e^{-\beta(t-t_j)} + \sum_{t_i^O<t} \alpha^O e^{-\beta(t-t_i^O)})).$$

By observing similar system dynamics to our base model, we can simplify the objective function and derive the HJB equation by letting $x_t = \sum_{t_j<t} \alpha e^{-\beta(t-t_j)} + \sum_{t_i^O<t} \alpha^O e^{-\beta(t-t_i^O)}$. The state variable x_t can be viewed as the excitation level of arrived customers, including both purchasers and non-purchasers. Unlike the excitation level of purchasing customers in the base model, x in this extension tends to be larger. Adopting the intensity λ as the decision variable, we obtain

$$\begin{aligned} &-\frac{\partial}{\partial t}V_t(x) + x\beta\frac{\partial}{\partial x}V_t(x) \\ &\quad = (h(x)+\varphi)\left(\max_{\lambda\in[\underline{\lambda},\bar{\lambda}]}\{r(\lambda)+\lambda(V_t(x+\alpha)-V_t(x+\alpha^O))\} + V_t(x+\alpha^O) - V_t(x)\right). \end{aligned} \tag{5}$$

In the right-hand side of (5), $V_t(x+\alpha) - V_t(x+\alpha^O)$ represents the future impact of currently having a purchasing customer as opposed to a non-purchasing customer, while $V_t(x+\alpha^O) - V_t(x)$ represents the future impact of currently having a non-purchasing customer as opposed to no arrival. The optimal intensity is determined to balance the current period profit with the future profit increment from one more purchase.

It is also worth noting that when $\alpha = \alpha^O$, the future additional benefit of currently having a purchasing customer as opposed to a non-purchasing customer is zero. Hence, there is no need to sacrifice the current profit margin to induce purchases and stimulate future demand. The seller would then optimize the price to achieve the highest profit greedily for the current period, resulting in a constant optimal price. In this scenario, the arrival process is independent of prices, and customer arrivals follow a self-exciting process with constant excitation intensity. Let $\lambda^{O*}(x)$ denote the optimal intensity for the general spreading model. We therefore obtain the following result.

LEMMA 3. *If $\alpha^O = \alpha$, the optimal intensity is constant, $\lambda^{O*}(x) = \arg\max_{0\le\lambda\le\lambda(0)}\{r(\lambda)\}$. If $\alpha^O = 0$, the optimal intensity is the one from the base model, $\lambda^{O*}(x) = \lambda^*(x)$.*

Lemma 3 presents two extreme cases for α^O. If purchasing and non-purchasing customers excite new customers with the same intensity, the optimal intensity aligns with that of a Poisson arrival process. Conversely, if non-purchasing customers excite zero new customers, the general information spreading model reduces to our base model. From this point forward, we assume $0 < \alpha^O < \alpha$. It can be easily verified that the value function $V_t(x)$ satisfies Propositions 2 and 3. Additionally, Theorems 1 and 2 can be verified through similar proofs. Denote by $V_t(x;\alpha^O)$ the expected total optimal profit for a general α^O.

PROPOSITION 6. *For $t \in [0,T]$, $V_t(x;\alpha^O)$ increases in α^O.*

Proposition 6 demonstrates that as α^O increases, the optimal profit also increases. We previously discussed that when $\alpha^O = 0$, the problem reduces to our base model. Since the value function increases with α^O, we can conclude that the optimal value for the general information spreading model is higher than that of our base model.

We next investigate whether the optimal intensity is monotone in α^O. It might be expected that the optimal intensity decreases in α^O. This expectation is supported by Lemma 3 showing that the optimal intensity is the Poisson arrival intensity if $\alpha^O = \alpha$ and the optimal intensity corresponds to the base model's optimal intensity when $\alpha^O = 0$, which is greater than the Poisson arrival intensity. By examining (5), we can calculate the optimal intensity as follows:

$$\lambda_t^{O*}(x) = \arg\max_{0\le\lambda\le\lambda(0)}\{r(\lambda) + \lambda(V_t(x+\alpha;\alpha^O) - V_t(x+\alpha^O;\alpha^O))\}.$$

Along with the concavity of $r(\lambda)$, we obtain that $\lambda_t^{O*}(x)$ satisfies

$$\lambda p'(\lambda) + p(\lambda) - c = -(V_t(x+\alpha;\alpha^O) - V_t(x+\alpha^O;\alpha^O)).$$

There are opposing effects in the right-hand side of the above equation, specifically in the term $V_t(x+\alpha;\alpha^O) - V_t(x+\alpha^O;\alpha^O)$. The interaction between x and α^O might lead to an increasing trend as α^O rises, which can be interpreted as a future increase in the number of non-purchasing customers. However, since $V_t(x;\alpha^O)$ increases with x, an increase in α^O would reduce the cost difference $V_t(x+\alpha;\alpha^O) - V_t(x+\alpha^O;\alpha^O)$, which can be seen as a future benefit reduction in the disparity between currently having a purchasing vs. non-purchasing customer. Due to these opposing effects, the optimal intensity $\lambda_t^{O*}(x)$ may not be monotonic as a function of α^O.

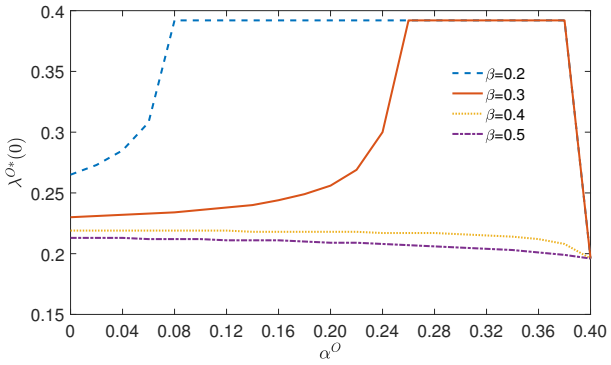


Figure 4 **Optimal intensities under different α^O**

To illustrate this non-monotonicity, we provide a numerical example. Suppose $\alpha = 0.4$, $\lambda(p) = 0.392(1-p)$, and $V_T(x) = x$. We compute the optimal intensity for $x_0 = 0$ and $T = 600$, considering scenarios with $\beta \in 0.2, 0.3, 0.4, 0.5$. Figure 4 displays that the optimal intensities decrease when the decay rate is high, specifically for $\beta = 0.4$ and $\beta = 0.5$. In these scenarios, customers are more forgetful, making the non-purchasing customer less critical, which leads to a decrease in the optimal intensities. Conversely, for lower decay rates, such as $\beta = 0.2$ and $\beta = 0.3$, the optimal intensities initially increase before decreasing. Here, customers are less forgetful, and the future increase in non-purchasing customers causes the optimal intensities to rise initially. Additionally, the optimal intensities are capped by an upper bound under the linear demand function. It is also notable that regardless of the value of β, the optimal intensity for $\alpha^O = \alpha = 0.4$ aligns with the Poisson arrival intensity, causing the optimal intensities to decrease around $\alpha = 0.4$.

5.2. Discrete Price Case

In this subsection, we assume that the seller offers a finite set of exogenously determined discrete prices denoted by $\{p_1, p_2, \ldots, p_K\}$ with $p_1 < p_2 < \cdots < p_K$ and corresponding purchase probabilities $\lambda_1 > \lambda_2 > \cdots > \lambda_K$. Given that the optimal purchase probability is greater than the Poisson arrival intensity λ^0, we assume that $\lambda_1 > \lambda_2 > \cdots > \lambda_K \geq \lambda^0$. The objective function is updated as follows:

$$-\frac{\partial}{\partial t}V_t(x) + x\beta\frac{\partial}{\partial x}V_t(x) = (h(x)+\varphi)\max_{\lambda\in\{\lambda_1,\ldots,\lambda_K\}}\{r(\lambda) + \lambda(V_t(x+\alpha) - V_t(x))\}.$$

It can be easily verified that the value function $V_t(x)$ satisfies parts (a), (b), and (c) of Proposition 2 and Theorem 2(a). Furthermore, we can derive the optimal intensity in the following theorem.

THEOREM 5. *For $t \in [0,T]$:*

(a) If $h(x)$ is convex, there exists $0 \leq R_{t,1} \leq R_{t,2} \leq \cdots \leq R_{t,K-1}$, such that $\lambda_t^*(x) = \lambda_1$ if $x < R_{t,1}$; $\lambda_t^*(x) = \lambda_i$ if $R_{t,i-1} \leq x < R_{t,i}$, for $i = 2, \ldots, K-1$; $\lambda_t^*(x) = \lambda_K$, otherwise.

(b) If $h(x)$ is linear, there exists $0 \leq \hat{T}_1 \leq \hat{T}_2 \leq \cdots \leq \hat{T}_{K-1} \leq T$, such that $\lambda_t^*(x) = \lambda_1$ if $t < \hat{T}_1$; $\lambda_t^*(x) = \lambda_i$ if $\hat{T}_{i-1} \leq t < \hat{T}_i$, for $i = 2, \ldots, K-1$; $\lambda_t^*(x) = \lambda_K$, otherwise.

Theorem 5 demonstrates that, on the one hand, with a linear self-exciting function, the optimal intensity is constant within each period regardless of the excitement level and increases across periods. On the other hand, with a convex self-exciting function, there are $K-1$ thresholds on the excitement level for each time, resulting in the optimal intensity being the same when the excitement level falls between the two consecutive thresholds. This raises a pertinent question: can the discrete price heuristic be asymptotically optimal? Somewhat surprisingly, we will show that the finitely discrete price heuristic is not asymptotically optimal in the following theorem.

THEOREM 6. *Consider any heuristic u where the available price set is finite. The heuristic is not asymptotically optimal, i.e., $V^*(n) - V^u(n) = \Theta(n)$.*

Theorem 6 demonstrates that the discrete price heuristic is not asymptotically optimal. This finding is both intriguing and distinct from the RM literature. From Theorem 4, for any heuristic to be asymptotically optimal, the expected purchase probability and excitement level must be "close" to the optimal solution in the deterministic problem. Unlike traditional RM problems where that solution is fixed, the optimal deterministic solution in our problem is nonstationary in time. Even though it may be possible to match the expected excitement level by using discrete prices, doing so would lead to sub-optimal profit due to the concavity of the profit rate function $r(\cdot)$. Since the discrete price heuristic is not asymptotically optimal, pricing managers had better avoid relying solely on a fixed set of discrete prices. Instead, they are better off constantly monitoring market trends and customer behavior and updating prices accordingly without confining themselves to the given discrete price set. This responsiveness to market changes will help set prices that are more closely aligned with the optimal pricing policy under no constraint of having to select from a predetermined set of discrete prices.

5.3. Multiple Products

In practice, a seller may offer a number of products in the same category that only differ in size and style. One purchase of any product can excite future customers who may end up purchasing other products in the same category. In this subsection, we assume that there are M differentiated products. A common stream of customers is shared by all products, with the customer arrival rate at time t modeled as a multivariate stochastic point process. Specifically, we assume that it follows a multivariate self-exciting process, defined as follows: $\Phi_t(\mathcal{H}_t) = \varphi + h(\sum_{n=1}^{M} \sum_{t_i^n < t} \alpha e^{-\beta(t-t_i^n)})$, where $\alpha \geq 0$, $\beta \geq 0$, $\varphi > 0$ and $h(\cdot)$ is a positive and increasing function. The price vector of those products is denoted by $\mathbf{p} = (p^1, p^2, \ldots, p^M)$. Upon a customer arrival, her purchasing probability of product $m = 1, 2, \ldots, M$ is denoted by $\lambda_t^m(\mathbf{p})$, determined by the current price vector $\mathbf{p}$. For each unit of sales of product $m = 1, 2, \ldots, M$, the seller incurs a marginal cost of c^m. We thereafter write down the purchase rate as follows: $\Lambda_t^m(\mathbf{p}) = \lambda_t^m(\mathbf{p})(\varphi + h(\sum_{n=1}^{M} \sum_{t_i^n < t} \alpha e^{-\beta(t-t_i^n)}))$. Following the conventions in RM (e.g., Gallego and Van Ryzin 1994), we use the purchase probabilities $\boldsymbol{\lambda} = (\lambda^1, \lambda^2, \ldots, \lambda^M)$ as the decision variables and make similar standard assumptions on the profit rate functions $r^m(\boldsymbol{\lambda}) = (p^m(\boldsymbol{\lambda}) - c^m)\lambda^m$ as in our base model.

By defining $x := \sum_{n=1}^{M} \sum_{t_i^n < t} \alpha e^{-\beta(t-t_i^n)}$, following the analogous steps from Proposition 1, we can express the seller's optimization problem as follows:

$$-\frac{\partial}{\partial t} V_t(x) + x\beta \frac{\partial}{\partial x} V_t(x) = (h(x) + \varphi) \max_{\boldsymbol{\lambda}_t = (\lambda_t^1, \lambda_t^2, \ldots, \lambda_t^M)} \{ \sum_{m=1}^{M} (r^m(\boldsymbol{\lambda}_t) + \lambda_t^m (V_t(x+\alpha) - V_t(x))) \}. \quad (6)$$

Even with multiple products that contribute to exciting customers, the system dynamics are similar to those of a single product, as we can capture the system state by a single-dimensional exciting level x. Consequently, we anticipate that the seller's value function $V_t(x)$ has properties that align with Proposition 2. Moreover, by applying the first-order condition to the maximization problem on the right-hand side of (6), we obtain that at the optimal demand rates $\boldsymbol{\lambda}_t^*$, the marginal profit rates across products are the same, i.e., $\partial r^1(\boldsymbol{\lambda})/\partial\lambda^1|_{\boldsymbol{\lambda}=\boldsymbol{\lambda}_t^*} = \partial r^2(\boldsymbol{\lambda})/\partial\lambda^2|_{\boldsymbol{\lambda}=\boldsymbol{\lambda}_t^*} = \cdots = \partial r^M(\boldsymbol{\lambda})/\partial\lambda^M|_{\boldsymbol{\lambda}=\boldsymbol{\lambda}_t^*}$. This implies that for a given t and x, the seller would use one product with a lower marginal cost as a "loss leader" by keeping its price lower to excite future customer arrivals who may end up purchasing other products with a higher margin.

While we omit the proofs due to repetition, all the asymptotic properties in Section 4 hold under the multiple products setup when we scale up the intensity by n. As the dynamics share a similar form with that of a single product in Equation (3), it can be shown that the optimal purchase probability for each product λ_t^m changes "slowly" at the rate of $1/n$ with respect to the excitement level x, the same rate as our single product base model. This indicates that the variances of λ_t^m are "small," which serves as the basis of all asymptotic results.

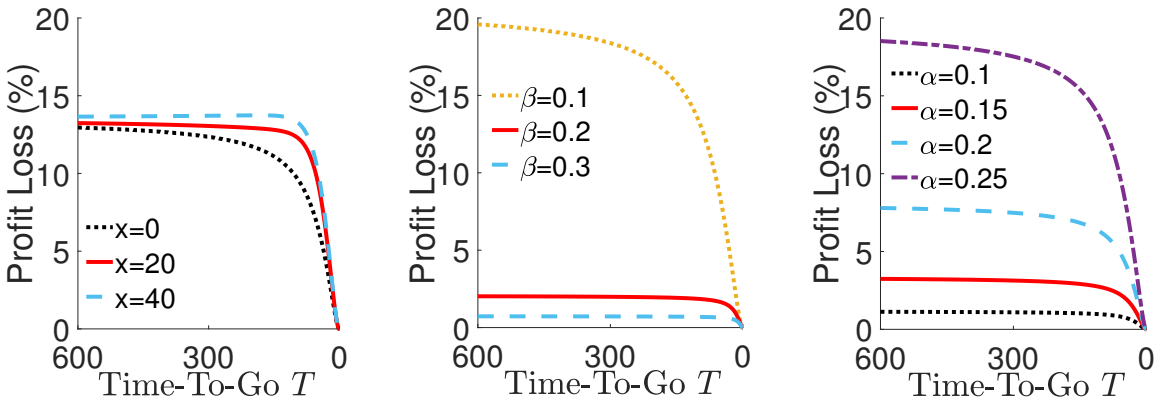
6. Numerical Study

In this section, we propose various easy-to-implement heuristics for the base model to derive further insights into the seller's heuristic decision-making processes. Sections 6.1 and 6.2 evaluate the impact on profit margins when neglecting the customers' self-exciting nature and implementing the static pricing, respectively. Section 6.3 investigates the performances of the deterministic heuristics.

Our base numerical setting is largely the same as our motivation example. We analyze sunscreen products on March 8th from JD.com. Using the maximum likelihood method, We identify a self-exciting process with parameters $\varphi = 0.392$, $\tilde{\alpha} = 0.09$, and $\beta = 0.11$. We assume that the customer valuations are scaled and uniformly distributed in $[0, 1]$. The organic arrival intensity for customers with independent needs is normalized to 1, and the excitation intensity is calculated as $\alpha = \tilde{\alpha}/\varphi = 0.091/0.392 \approx 0.23$. Given that the item was sold over a period of approximately 10 hours, we set the time horizon to roughly 10 hours, i.e., $T = 600$ minutes. We normalize the marginal cost to zero and adopt the self-exciting function as $h(x) = x + 0.001x^2$. To achieve convergence of the optimal solution in our computations, we use the Semi-Lagrangian method to solve the Hamilton-Jacobi-Bellman (HJB) equation. We discretize x with a step size of 0.01 and time t with a step size of 0.1. Linear interpolation is applied when the calculated values do not align with the lattice points. Unless otherwise specified, the parameters utilized in this section adhere to the aforementioned base numerical settings.

6.1. Profit Loss of Myopic Pricing

In this subsection, we examine the profit loss of myopically maximizing the profit rate for a given customer arrival and neglecting this customer's future impact of exciting other customers. When the seller ignores the self-excitement of the arrival process and considers only organic customers, the intensity is based on a Poisson process for customer arrivals. In our base scenario, customer valuations are uniformly distributed within the interval $[0,1]$, and those arriving with independent needs follow the same distribution. Consequently, the seller maximizes the organic intensity $\lambda^0 = \max_{\lambda\geq 0} \lambda p(\lambda) = \max_{\lambda\geq 0} \lambda(1-\lambda/0.392) = 0.196$. This λ^0 represents the intensity of Poisson arrivals, serving as the lower bound for the optimal intensity. We denote the expected profit under the arrival intensity λ^0 as $V_t^{Myopic}(x)$ and define Profit Loss $\% = (V_t(x) - V_t^{Myopic}(x))/V_t(x) \times 100\%$. The following figures display the profit loss when neglecting customers' self-excitement.



(a) Initial Excitement Level (b) Decay Rate (c) Excitation Intensity

Figure 5 Percentage profit loss of Myopic Pricing

From Figure 5(a), we observe that ignoring self-excitement can result in a profit loss of up to roughly 13%. Furthermore, the profit loss increases as the time-to-go increases, eventually converging to an upper limit. Furthermore, we note that as the excitement level x increases, the profit loss slightly increases. This is due to the increasing property of the self-exciting function. As the excitement level x increases, the incremental benefit of currently having one purchase also increases.

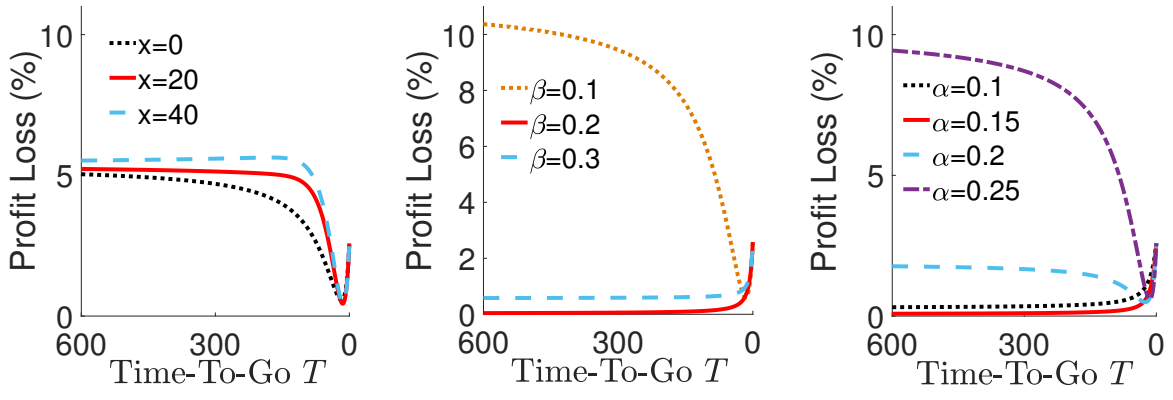
Figure 5(b) shows that the profit loss of ignoring self-excitement decreases as the decay rate β increases. Specifically, when $\beta = 0.1$, the profit loss is up to 20%, whereas when $\beta = 0.3$, the profit loss falls below 1%. This phenomenon occurs because a higher decay rate implies that customers are more forgetful and quickly cease to excite other customers. Consequently, the word-of-mouth effect diminishes, causing customer arrivals close to a Poisson process. Figure 5(c) illustrates that

the profit loss increases as α rises. For instance, when $\alpha = 0.25$, the profit loss reaches up to 18%. However, with $\beta = 0.15$, the profit gap remains below 3.2%. This indicates that as the excitation intensity decreases, the customers are less likely to excite other customers, reducing the word-of-mouth effect and thereby decreasing the profit loss of ignoring self-excitement. Consequently, lower excitation intensity results in less profit loss.

Based on the above observations, pricing managers are advised to employ advanced analytics tools to model and calibrate the self-excitement process, which can help them understand how current sales influence future demand. Additionally, they are better off by ensuring that their pricing policies take into account self-exciting behavior, such as the word-of-mouth effect, because ignoring these factors can result in significant profit losses.

6.2. Profit Loss of Static Pricing

Next, we consider an alternative heuristic policy where the seller determines the price statically based on the expected number of customer arrivals, including organic and self-excited customers. With 279 customers arriving over 613 minutes, as observed from the JD.com data, the average arrival rate is $279/613 \approx 0.455$. Assuming customers are uniformly distributed in the interval [0,1] as in our base numerical setting, the optimal static arrival rate is calculated as $\lambda^{Static} = 0.455/2 \approx 0.228$. We denote by $V_t^{Static}(x)$ the expected profit under the arrival intensity λ^{Static} and define Profit Loss $\% = (V_t(x) - V_t^{Static}(x))/V_t(x) \times 100\%$. As in the previous subsection, we can evaluate the performance of the static pricing heuristic.



(a) Initial Excitement Level (b) Decay Rate (c) Excitation Intensity

Figure 6 Percentage profit loss of Static Pricing

Figure 6(a) indicates that determining the price statically based on the expected number of customer arrivals results in an approximately 5% profit loss. However, as the remaining time

decreases, the profit loss initially decreases and then increases. This down-then-up behavior is driven by the observation of a higher average expected optimal intensity over time when there is more time in the sales horizon for any given initial excitement level. When the time-to-go is long enough, the average expected optimal intensity over the remaining time is higher than the static price but decreases and gets closer to the static price as the time-to-go becomes shorter, leading to a reduction in profit loss of using the static price. Once the average expected optimal intensity over the remaining time falls below the static intensity and deviates further from it as the remaining time becomes shorter, the profit loss starts to increase.

Figure 6(b) illustrates that the profit loss from adopting static pricing does not necessarily follow a monotonic trend as the decay rate β increases. Specifically, when $\beta = 0.2$, the profit loss is below 2%, which is lower than the profit loss observed for both $\beta = 0.1$ and $\beta = 0.3$. This occurs because a higher decay rate indicates that customers quickly lose interest and stop influencing others, reducing the word-of-mouth effect and causing a decrease in the optimal intensity. The average expected optimal intensity for $\beta = 0.2$ is closer to the static intensity compared to $\beta = 0.1$ and $\beta = 0.3$. As a result, the performance for $\beta = 0.2$ surpasses that of both $\beta = 0.1$ and $\beta = 0.3$. Figure 6(c) shows that the profit loss from static pricing does not necessarily exhibit a monotonic trend as the excitation intensity α increases. For example, the profit loss for $\alpha = 0.15$ is lower compared to $\alpha = 0.1$ and $\alpha = 0.2$ when the remaining time is long. This occurs because as the excitation intensity decreases, customers are less likely to excite others, diminishing the word-of-mouth effect and causing a decrease in the average expected optimal intensity over the remaining time. The expected optimal intensity for $\alpha = 0.15$ is closer to the static intensity than $\alpha = 0.1$ and $\alpha = 0.2$, resulting in better performance of adopting static pricing.

As shown above, the static pricing approach can lead to a significant profit loss. Adopting dynamic pricing can assist companies in maximizing profits by adjusting prices in response to real-time customer arrivals and information dissemination over social networks.

6.3. Profit Loss of Deterministic Heuristic

In this subsection, we adopt the optimal intensity over time from the deterministic problem, referred to as the deterministic heuristic. By solving the associated ordinary differential equations, we find that the deterministic intensity over time depends solely on time and can be computed numerically using the standard forward-backward sweep algorithm. The time interval $[0, T]$ is discretized with a step size δ. In each iteration, we update x_t, λ_t, and μ_t according to the first-order Hamiltonian conditions. The algorithm stops when x_t, λ_t, and μ_t all converge. Details are provided in Algorithm 1. We denote by $V_t^D(x)$ the expected profit under the arrival intensity λ^D and define Profit Loss $\% =$

Algorithm 1 Forward-Backward Sweep Algorithm for the Deterministic Problem

Initialize with $\lambda_t = \lambda^*$, $x_t = 0$ and $\mu_t = 0$ for any $t = 0, \delta, 2\delta, \cdots, T$.

Initiate $GAP = 1$

while $GAP \geq 0.001$ **do**

Calulate $x_{t+\delta}^{new} = x_t^{new} + \alpha n \lambda_t \left[1 + h\left(x_t^{new}/n\right)\right] - \beta x_t^{new}$ for $t = 0, \delta, 2\delta, \cdots, T - \delta$, with $x_0 = 0$.

Calulate $\mu_{t-\delta}^{new} = \mu_t^{new} + r(\lambda_t) h'\left(x_t^{new}/n\right) + \left[\alpha \lambda h'\left(x_t^{new}/n\right) - \beta\right] \mu_t^{new}$ for $t = T, T - \delta, \cdots \delta$, with $\mu_T^{new} = 0$.

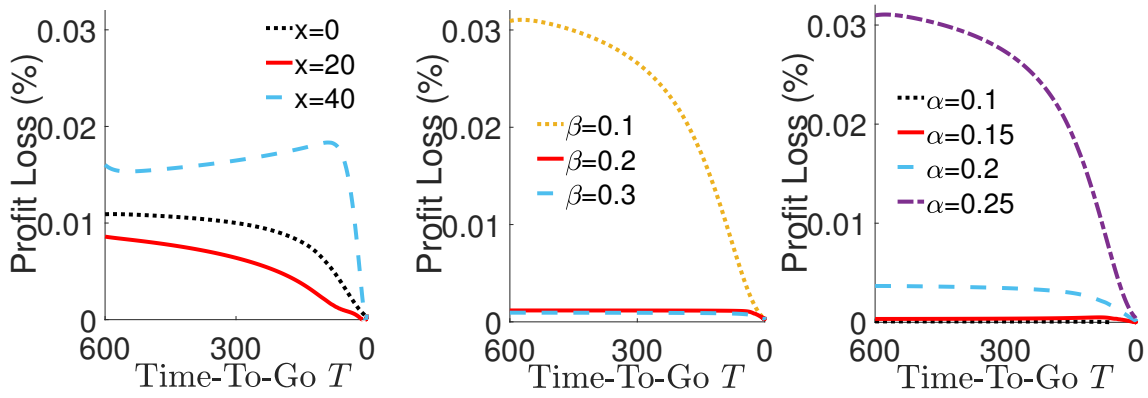
Calculate λ_t^{new} such that $r'(\lambda_t^{new}) + \alpha \mu_t^{new} = 0$ for $t = 0, \delta, 2\delta, \cdots, T$.

Update $GAP = \max\left\{\sum_t |\lambda_t - \lambda_t^{new}| / \sum_t \lambda_t^{new}, \sum_t |x_t - x_t^{new}| / \sum_t x_t^{new}, \sum_t |\mu_t - \mu_t^{new}| / \sum_t \mu_t^{new}\right\}$.

Update $\lambda_t = \lambda_t^{new}$, $x_t = x_t^{new}$ and $\mu_t = \mu_t^{new}$.

end while

$(V_t(x) - V_t^D(x))/V_t(x) \times 100\%$. As in the previous subsection, we can evaluate the performance of the deterministic pricing heuristic.



(a) Initial Excitement Level (b) Decay Rate (c) Excitation Intensity

Figure 7 Percentage profit loss of Deterministic Heuristic

Figure 7(a) demonstrates that the time-varying deterministic heuristic significantly outperforms both myopic and static pricing heuristics because it can, to some degree, capture the time-varying nature of the optimal price trajectories (see, e.g., Theorem 3) while both myopic and static pricing cannot. The maximum profit loss is less than 0.02%, 0.012%, and 0.009% for initial excitement levels $x = 0$, $x = 20$, and $x = 40$, respectively. Figure 7(b) shows that the profit loss of the deterministic heuristic decreases as the decay rate increases. A higher decay rate means that customers quickly lose interest and stop influencing others, diminishing the word-of-mouth effect and shifting the

customer arrival process closer to a Poisson process, which has a lower variability than a general Hawkes process driven by self-excitement. Consequently, it makes the deterministic heuristic closer to the (random) optimal intensity of the stochastic problem that has less variability to react to. Figure 7(c) indicates that the profit loss of the deterministic heuristic increases as the excitation intensity increases. When the excitation intensity decreases, customers are less likely to excite others, reducing the word-of-mouth effect and shifting the customer arrival process closer to a Poisson process. For the same reason mentioned above, lower excitation intensity results in less profit loss when adopting deterministic heuristics.

Our numerical experiments confirm the deterministic heuristic's superior performance, which is theoretically demonstrated in Theorem 4. The deterministic heuristic simplifies the pricing decision-making process, allowing pricing managers to make well-performed pricing decisions efficiently.

7. Conclusion

In this paper, we investigate how a seller can dynamically adjust selling prices under a generalized self-exciting customer arrivals model, where each purchase stimulates additional purchases, increasing the arrival intensity. Self-excitement complicates the optimal pricing policy. We identify the excitement level as the key sufficient statistic and fully characterize the structural properties of the optimal policy. Our findings indicate that the optimal pricing policy encourages higher purchase rates as the excitement level rises and that the optimal price trajectory exhibits an upward trend in the absence of decay. Moreover, we examine the deterministic version of our problem. We show that the performance loss of the deterministic time-varying (open-loop) heuristic from the optimal profit is just a constant. We also extend our base model to account for general information spreading, discrete pricing, and multiple products, offering comprehensive insights for dynamic pricing strategies in various market conditions. Finally, our numerical results highlight the critical value of accounting for self-excitement in pricing decisions because neglecting this factor leads to significant profit loss.

To our knowledge, this is the first study to model customer arrivals as a self-exciting process in RM, capturing how past purchases influence future customer arrivals. Our findings reveal that the optimal price trajectory tends to increase over time, which fundamentally differs from the existing RM literature, where the optimal price tends to stay constant without capacity constraints. Perhaps surprisingly, we find that how the optimal price reacts to the excitement level depends on whether the market is in a growth or maturity phase. Additionally, we adopt the deterministic heuristic to the stochastic problem and demonstrate that its performance loss is a constant, which contrasts with traditional RM problems.

We point out some directions for future research. The first direction involves introducing the capacity constraint, a feature of the classic RM problems. The second direction is to consider separate streams of organic customers, each for one product in a category, but one product's purchase may excite arrivals for other products.

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Online Appendix to "Dynamic Pricing Under Self-Exciting Arrival Processes":

A. Major Proofs

Proof of Lemma 1. For a small $\delta > 0$, $\mathbb{P}(N_{t+\delta} - N_t > 1) = o(\delta)$. If $N_{t+\delta} - N_t = 0$,

$$x_{t+\delta} = \sum_{t_i < t+\delta} \alpha e^{-\beta(t+\delta-t_i)} = e^{-\beta\delta} x_t = x_t - \beta\delta x_t + o(\delta).$$

If $N_{t+\delta} - N_t = 1$, the arrival time $t_{N_{t+\delta}} = t + \delta$.

$$x_{t+\delta} = \sum_{i=0}^{N_t} \alpha e^{-\beta(t+\delta-t_i)} + \alpha e^{\beta\left(t+\delta t_{N_{t+\delta}}\right)} = e^{-\beta\delta} x_t + \alpha = x_t - \beta\delta x_t + \alpha + o(\delta). \quad \square$$

Proof of Proposition 1. From Lemma 1, $x_{t+\delta} = x_t - \beta x_t \delta$ with probability $1 - \lambda(h(x) + \varphi)\delta$. $x_{t+\delta} = x_t + \alpha - \beta x_t \delta$ with probability $\lambda(h(x) + \varphi)\delta$. $\mathbb{P}(x_{t+\delta} > x_t + \alpha - \beta x_t \delta) = o(\delta)$. Therefore,

$$\begin{aligned} V_t(x) = \max_{\lambda} \{ & (h(x) + \varphi) r(\lambda)\delta + \lambda(h(x) + \varphi)\delta V_{t+\delta}(x - \beta\delta x + \alpha(1 - \beta\delta)) \\ & + (1 - \lambda(h(x) + \varphi)\delta)[V_{t+\delta}(x - \beta\delta x)] \} + o(\delta). \end{aligned} \tag{A.1}$$

Rearranging and taking the limit as $\delta \to 0$, we get the announced result. $\square$

Proof of Proposition 2. (i) We prove by induction. The statement holds for $t = T$ as $V_T(x) = 0$. We assume $V_t(x)$ increases in x for some $t \le T$. Then for any $x^1 > x^2 \ge 0$, $V_t(x^1 - \beta\delta x^1) \le V_t(x^2 - \beta\delta x^2)$. $V_t(x^1 + \alpha) \le V_t(x^2 + \alpha)$. From the HJB equation (A.1),

$$\begin{aligned} V_{t-\delta}(x^1) &= \max_{\lambda} \left\{ \lambda(h(x^1) + \varphi)\delta \left[p(\lambda) - c + V_t(x^1 + \alpha) \right] + (1 - \lambda(h(x^1) + \varphi)\delta) V_t(x^1 - \beta x^1 \delta) \right\} + o(\delta) \\ &\ge \max_{\lambda} \left\{ \lambda(h(x^1) + \varphi)\delta \left[p(\lambda) - c + V_t(x^2 + \alpha) \right] + (1 - \lambda(h(x^1) + \varphi)\delta) V_t(x^2 - \beta x^2 \delta) \right\} + o(\delta). \end{aligned}$$

Since $h(x)$ strictly increases in x, $h(x^1) > h(x^2)$. Along with $V_t(x^2 + \alpha) \ge V_t(x^2 - \beta x^2 \delta)$, we have

$$\begin{aligned} V_{t-\delta}(x^1) &> \max_{\lambda} \left\{ \lambda(h(x^2) + \varphi)\delta \left[p(\lambda) - c + V_t(x^2 + \alpha) \right] + (1 - \lambda(h(x^2) + \varphi)\delta) V_t(x^2 - \beta x^2 \delta) \right\} + o(\delta) \\ &= V_{t-\delta}(x^2). \end{aligned}$$

Therefore, $V_{t-\delta}(x)$ strictly increases in x.

(ii). Denote $V_t^{\alpha}(x)$ as the expected total profit for arrival intensity α. We prove that for any $0 \le \alpha_1 < \alpha_2$ and $t < T$, $V_t^{\alpha_1}(x) < V_t^{\alpha_2}(x)$ by induction. Assume that the result holds for time $t < T$. Then $V_t^{\alpha_1}(x + \alpha_1) \le V_t^{\alpha_2}(x + \alpha_1)$ and $V_t^{\alpha_1}(x - \beta x\delta) \le V_t^{\alpha_2}(x - \beta x\delta)$. From the HJB equation,

$$V_{t-\delta}^{\alpha_1}(x) = \max_{\lambda} \{ (h(x) + \varphi)\delta r(\lambda) + \lambda(h(x) + \varphi)\delta V_t^{\alpha_1}(x + \alpha_1) + (1 - \lambda(h(x) + \varphi)\delta) V_t^{\alpha_1}(x - \beta x\delta) \} + o(\delta)$$

$$\leq \max_{\lambda}\{(h(x)+\varphi)\delta r(\lambda)+\lambda(h(x)+\varphi)\delta V_t^{\alpha_2}(x+\alpha_1)+(1-\lambda(h(x)+\varphi)\delta)V_t^{\alpha_2}(x-\beta x\delta)\}+o(\delta).$$

Since $V_t(x)$ strictly increases in x according to (i), $V_t^{\alpha_2}(x+\alpha_1)<V_t^{\alpha_2}(x+\alpha_2)$. Thus,

$$\begin{aligned}V_{t-\delta}^{\alpha_1}(x) &< \max_{\lambda}\{(h(x)+\varphi)\delta r(\lambda)+\lambda(h(x)+\varphi)\delta V_t^{\alpha_2}(x+\alpha_2)+(1-\lambda(h(x)+\varphi)\delta)V_t^{\alpha_2}(x-\beta x\delta)\}+o(\delta)\\ &= V_{t-\delta}^{\alpha_2}(x).\end{aligned}$$

Similarly, we denote $V_t^{\beta}(x)$ as the expected total profit with decay rate β. We then prove that for any $t<T$ and $0\leq\beta_1<\beta_2$, $V_t^{\beta_1}(x)<V_t^{\beta_2}(x)$ by induction. It is easy to show that the result holds for time T. Assume that the result holds for time t. Then $V_t^{\beta_1}(x+\alpha)\geq V_t^{\beta_2}(x+\alpha)$ and $V_t^{\beta_1}(x-\beta_1 x\delta)\geq V_t^{\beta_2}(x-\beta_1 x\delta)$. From the HJB equation,

$$\begin{aligned}V_{t-\delta}^{\beta_1}(x) &= \max_{\lambda}\left\{\lambda(h(x)+\varphi)\delta\left[p(\lambda)-c+V_t^{\beta_1}(x+\alpha)\right]+(1-\lambda(h(x)+\varphi)\delta)V_t^{\beta_1}(x-\beta_1 x\delta)\right\}+o(\delta)\\ &\geq \max_{\lambda}\left\{\lambda(h(x)+\varphi)\delta\left[p(\lambda)-c+V_t^{\beta_2}(x+\alpha)\right]+(1-\lambda(h(x)+\varphi)\delta)\,V_t^{\beta_2}(x-\beta_1 x\delta)\right\}+o(\delta).\end{aligned}$$

Since $V_t(x)$ strcitly increases in x according to (i), $V_t^{\beta_2}(x-\beta_1 x\delta)>V_t^{\beta_2}(x-\beta_2 x\delta)$. Thus,

$$\begin{aligned}V_{t-\delta}^{\beta_1}(x) &> \max_{\lambda}\left\{(h(x)+\varphi)\delta r(\lambda)+\lambda(h(x)+\varphi)\delta V_t^{\beta_2}(x+\alpha)+(1-\lambda(h(x)+\varphi)\delta)V_t^{\beta_2}(x-\beta_2 x\delta)\right\}+o(\delta)\\ &= V_{t-\delta}^{\beta_2}(x).\quad\square\end{aligned}$$

Proof of Proposition 3. To prove this proposition, we first prove the following lemma:

LEMMA 4. *If the profit rate function $r(\lambda)=\lambda(p(\lambda)-c)$ is strictly concave, $p(a)-p(b)+\frac{a}{b}(b-a)p'(a)>0$ for any $0\leq a<b$.*

Proof of Lemma. Denote $F(a)=p(a)-p(b)+\frac{a}{b}(b-a)p'(a)=p(a)+ap'(a)-p(b)-\frac{a^2}{b}p'(a)$. Take derivative w.r.t. a:

$$\begin{aligned}F'(a) &= p'(a)+p'(a)+ap''(a)-\frac{2a}{b}p'(a)-\frac{a^2}{b}p''(a)\\ &= \left(2-\frac{2a}{b}\right)p'(a)+a\left(1-\frac{a}{b}\right)p''(a)\\ &= \left(1-\frac{a}{b}\right)\left[2p'(a)+ap''(a)\right].\end{aligned}$$

Since $r(\lambda)=\lambda(p(\lambda)-c)$ is strictly concave, $2p'(\lambda)+\lambda p''(\lambda)<0$. Therefore, $F(a)$ decreases in a when $a\leq b$. Since $F(b)=0$, $F(a)>0$ for any $a<b$. $\square$

Now suppose the statement in Proposition 3 doesn't hold for $t\in[0,T]$. Then there exists $x_1<x_2$ such that $\Lambda_t^*(x_1)>\Lambda_t^*(x_2)$. Let $\frac{\Lambda_t^*(x_2)}{\varphi+h(x_2)}$ be the largest optimizer. Then

$V_t(x_2)$

$$= (\varphi + h(x_2))r\left(\frac{\Lambda_t^*(x_2)}{\varphi + h(x_2)}\right)\delta + \Lambda_t^*(x_2)\delta V_{t+\delta}\left(x_2(1-\beta\delta) + \alpha(1-\beta\delta_1)\right) + \left(1 - \Lambda_t^*(x_2)\delta\right)V_{t+\delta}\left(x_2(1-\beta\delta)\right)$$
$$= (\varphi + h(x_2))r\left(\frac{\Lambda_t^*(x_2)}{\varphi + h(x_2)}\right)\delta + \Lambda_t^*(x_1)\delta V_{t+\delta}\left(x_2(1-\beta\delta) + \alpha(1-\beta\delta_1)\right) + \left(1 - \Lambda_t^*(x_1)\delta\right)V_{t+\delta}\left(x_2(1-\beta\delta)\right)$$
$$+ \left(\Lambda_t^*(x_2) - \Lambda_t^*(x_1)\right)\delta\left[V_{t+\delta}\left(x_2(1-\beta\delta) + \alpha(1-\beta\delta_1)\right) - V_{t+\delta}\left(x_2(1-\beta\delta)\right)\right].$$

From Equation (3),

$$p\left(\frac{\Lambda_t^*(x_2)}{\varphi + h(x_2)}\right) - c + \frac{\Lambda_t^*(x_2)}{\varphi + h(x_2)}p'\left(\frac{\Lambda_t^*(x_2)}{\varphi + h(x_2)}\right) + V_{t+\delta}\left(x_2(1-\beta\delta) + \alpha(1-\beta\delta_1)\right) - V_{t+\delta}\left(x_2(1-\beta\delta)\right) = 0.$$

Thus,

$$V_t(x_2) = \Lambda_t^*(x_1)\delta V_{t+\delta}\left(x_2(1-\beta\delta) + \alpha(1-\beta\delta_1)\right) + \left(1 - \Lambda_t^*(x_1)\delta\right)V_{t+\delta}\left(x_2(1-\beta\delta)\right)$$
$$+(\varphi + h(x_2))r\left(\frac{\Lambda_t^*(x_2)}{\varphi + h(x_2)}\right)\delta - \left(\Lambda_t^*(x_2) - \Lambda_t^*(x_1)\right)\delta\left[p\left(\frac{\Lambda_t^*(x_2)}{\varphi + h(x_2)}\right) + \frac{\Lambda_t^*(x_2)}{\varphi + h(x_2)}p'\left(\frac{\Lambda_t^*(x_2)}{\varphi + h(x_2)}\right)\right]$$
$$= (\varphi + h(x_1))r\left(\frac{\Lambda_t^*(x_1)}{\varphi + h(x_2)}\right)\delta + \Lambda_t^*(x_1)\delta V_{t+\delta}\left(x_2(1-\beta\delta) + \alpha(1-\beta\delta_1)\right) + \left(1 - \Lambda_t^*(x_1)\delta\right)V_{t+\delta}\left(x_2(1-\beta\delta)\right)$$
$$+\delta\left[\Lambda_t^*(x_1)\left(p\left(\frac{\Lambda_t^*(x_2)}{\varphi + h(x_2)}\right) - p\left(\frac{\Lambda_t^*(x_1)}{\varphi + h(x_2)}\right)\right) + \Lambda_t^*(x_2)\left(\frac{\Lambda_t^*(x_1)}{\varphi + h(x_2)} - \frac{\Lambda_t^*(x_2)}{\varphi + h(x_2)}\right)p'\left(\frac{\Lambda_t^*(x_2)}{\varphi + h(x_2)}\right)\right].$$

from our assumption, $\Lambda_t^*(x_1) > \Lambda_t^*(x_2)$. Thus from Lemma 4,

$$\Lambda_t^*(x_1)\left(p\left(\frac{\Lambda_t^*(x_2)}{\varphi + h(x_2)}\right) - p\left(\frac{\Lambda_t^*(x_1)}{\varphi + h(x_2)}\right)\right) + \Lambda_t^*(x_2)\left(\frac{\Lambda_t^*(x_1)}{\varphi + h(x_2)} - \frac{\Lambda_t^*(x_2)}{\varphi + h(x_2)}\right)p'\left(\frac{\Lambda_t^*(x_2)}{\varphi + h(x_2)}\right) < 0.$$

Therefore,

$$V_t(x_2) < (\varphi + h(x_1))r\left(\frac{\Lambda_t^*(x_1)}{\varphi + h(x_2)}\right)\delta + \Lambda_t^*(x_1)\delta V_{t+\delta}\left(x_2(1-\beta\delta) + \alpha(1-\beta\delta_1)\right) + \left(1 - \Lambda_t^*(x_1)\delta\right)V_{t+\delta}\left(x_2(1-\beta\delta)\right).$$

Note that the RHS is the expected profit if the seller chooses the rate $\frac{\Lambda_t^*(x_1)}{\varphi + h(x_2)}$ at time t when the excitement level is x_2. This leads to contradiction as $\frac{\Lambda_t^*(x_2)}{\varphi + h(x_2)}$ is the largest optimizer. □

Proof of Theorem 1. Since $r'(\lambda_t^*(x)) = V_t(x) - V_t(x+\alpha)$. It is sufficient to show that $\frac{\partial^2 V}{\partial t \partial x} \le 0$. Take derivative of Equation w.r.t. t:

$$-\frac{\partial^2 V}{\partial t^2} + \beta x\frac{\partial^2 V}{\partial t \partial x} = (\varphi + h(x))\left[r'(\lambda_t^*(x))\frac{\partial \lambda_t^*(x)}{\partial t} + \frac{\partial \lambda_t^*(x)}{\partial t}\left(V_t(x+\alpha) - V_t(x)\right) + \lambda_t^*(x)\frac{\partial}{\partial t}\left(V_t(x+\alpha) - V_t(x)\right)\right].$$

Note that $r'(\lambda_t^*(x)) + V_t(x+\alpha) - V_t(x) = 0$. Thus

$$-\frac{\partial^2 V}{\partial t^2} + \beta x\frac{\partial^2 V}{\partial t \partial x} = (\varphi + h(x))\lambda_t^*(x)\left[\frac{\partial}{\partial t}V_t(x+\alpha) - \frac{\partial}{\partial t}V_t(x)\right].$$

Let $f_t(x) = \frac{\partial V}{\partial t}$. We prove by induction that $f_t(x)$ decreases in x. When $t = T$, $f_t(x) = -r^0(\varphi + h(x))$ decreases in x. It is thereby sufficient to show that if $f_t(x)$ decreases in x, $f_{t-\delta}(x)$ strictly decreases in x for a small δ. Consider any $x_2 > x_1 \ge 0$:

$$f_{t-\delta}(x_1) = \left[1 - (\varphi + h(x_1))\lambda_{t-\delta}^*(x_1)\delta\right]f_t(x_1 - \beta\delta x_1\delta) + (\varphi + h(x_1))\lambda_{t-\delta}^*(x_1)\delta f_t(x_1 + \alpha) + o(\delta)$$

From induction assumption, $f_t(x_1-\beta\delta x_1\delta) \geq f_t(x_1+\alpha)$. From Proposition 3, $(\varphi+h(x_1))\lambda^*_{t-\delta}(x_1) < (\varphi+h(x_2))\lambda^*_{t-\delta}(x_2)$. Thus,

$$f_{t-\delta}(x_1) > \left[1-(\varphi+h(x_2))\lambda^*_{t-\delta}(x_2)\delta\right] f_t(x_1-\beta\delta x_1\delta) + (\varphi+h(x_2))\lambda^*_{t-\delta}(x_1)\delta f_t(x_1+\alpha) + o(\delta)$$

Again from induction assumption, $f_t(x_1-\beta\delta x_1\delta) \geq f_t(x_2-\beta\delta x_2\delta)$. $f_t(x_1+\alpha) \geq f_t(x_2+\alpha)$. Thus,

$$f_{t-\delta}(x_1) > \left[1-(\varphi+h(x_2))\lambda^*_{t-\delta}(x_2)\delta\right] f_t(x_2-\beta\delta x_2\delta) + (\varphi+h(x_2))\lambda^*_{t-\delta}(x_2)\delta f_t(x_2+\alpha) + o(\delta) = f_{t-\delta}(x_2). \quad \square$$

Proof of Theorem 2. Note that $\lambda^*_t(x)$ is the solution of Equation 3. It is therefore solved by $r'(\lambda) = V_t(x) - V_t(x+\alpha)$. Since $r(\lambda)$ is strictly concave, $r'(\lambda)$ strictly decreases in λ. Therefore, $\lambda^*_t(x)$ strictly increases (decreases) in x if $V_t(x)$ is strictly convex (concave) in x.

We first prove (i) that if $h(x)$ is strictly convex in x, $V_t(x)$ is strictly convex in x by induction. Since $V_T(x) = 0$ for any x, the result holds for time T. We assume that $V_{t+\delta}(x)$ is strictly convex in x for a small $\delta > 0$. Let $x^1 < x^2$, $\bar{x} = \theta x^1 + \bar{\theta} x^2$ where $0 < \theta < 1$, $\bar{\theta} = 1-\theta$. Since $V_{t+\delta}(x)$ is convex in x,

$$\begin{aligned}
&V_t(\bar{x}) = r(\lambda^*_t(\bar{x}))(h(\bar{x})+\varphi)\delta + \lambda^*_t(\bar{x})(h(\bar{x})+\varphi)\delta V_{t+\delta}\left(\bar{x}(1-\beta\delta)+\alpha(1-\beta\delta_1)\right) \\
&+ (1-\lambda^*_t(\bar{x})(h(\bar{x})+\varphi)\delta) V_{t+\delta}\left(\bar{x}(1-\beta\delta)\right) + o(\delta) \\
\leq& r(\lambda^*_t(\bar{x}))(h(\bar{x})+\varphi)\delta + \lambda^*_t(\bar{x})(h(\bar{x})+\varphi)\delta\left[\theta V_{t+\delta}\left(x^1(1-\beta\delta)+\alpha(1-\beta\delta_1)\right) + \bar{\theta} V_{t+\delta}\left(x^2(1-\beta\delta)+\alpha(1-\beta\delta_1)\right)\right] \\
&+ (1-\lambda^*_t(\bar{x})(h(\bar{x})+\varphi)\delta)\left[\theta V_{t+\delta}\left(x^1(1-\beta\delta)\right) + \bar{\theta} V_{t+\delta}\left(x^2(1-\beta\delta)\right)\right] + o(\delta) \\
=& r(\lambda^*_t(\bar{x}))(h(\bar{x})+\varphi)\delta + \theta V_{t+\delta}\left(x^1(1-\beta\delta)\right) + \bar{\theta} V_{t+\delta}\left(x^2(1-\beta\delta)\right) \\
&+ \theta\lambda^*_t(\bar{x})(h(\bar{x})+\varphi)\delta\Big\{\theta\left[V_{t+\delta}\left(x^1(1-\beta\delta)+\alpha(1-\beta\delta_1)\right) - V_{t+\delta}\left(x^1(1-\beta\delta)\right)\right] \\
&+ \bar{\theta}\left[V_{t+\delta}\left(x^2(1-\beta\delta)+\alpha(1-\beta\delta_1)\right) - V_{t+\delta}\left(x^2(1-\beta\delta)\right)\right]\Big\} + o(\delta).
\end{aligned}$$

Since $h(x)$ is strictly convex,

$$\begin{aligned}
&V_t(\bar{x}) < r(\lambda^*_t(\bar{x}))(\theta h(x^1) + \bar{\theta} h(x^2) + \varphi)\delta + \theta V_{t+\delta}\left(x^1(1-\beta\delta)\right) + \bar{\theta} V_{t+\delta}\left(x^2(1-\beta\delta)\right) \\
&+ \theta\lambda^*_t(\bar{x})(\theta h(x^1) + \bar{\theta} h(x^2) + \varphi)\delta\Big\{\theta\left[V_{t+\delta}\left(x^1(1-\beta\delta)+\alpha(1-\beta\delta_1)\right) - V_{t+\delta}\left(x^1(1-\beta\delta)\right)\right] \\
&+ \bar{\theta}\left[V_{t+\delta}\left(x^2(1-\beta\delta)+\alpha(1-\beta\delta_1)\right) - V_{t+\delta}\left(x^2(1-\beta\delta)\right)\right]\Big\} + o(\delta) \\
=& \theta V_{t+\delta}(x^1(1-\beta\delta)) + \bar{\theta} V_{t+\delta}(x^2(1-\beta\delta)) \\
&+ \left(\theta(h(x^1)+\varphi) + \bar{\theta}(h(x^2)+\varphi)\right)\delta\Big\{\theta\left[r\left(\lambda^*_t(\bar{x})\right) + \lambda^*_t(\bar{x})\left[V_{t+\delta}\left(x^1(1-\beta\delta)+\alpha(1-\beta\delta_1)\right) - V_{t+\delta}\left(x^1(1-\beta\delta)\right)\right]\right] \\
&+ \bar{\theta}\left[r\left(\lambda^*_t(\bar{x})\right) + \lambda^*_t(\bar{x})\left[V_{t+\delta}\left(x^2(1-\beta\delta)+\alpha(1-\beta\delta_1)\right) - V_{t+\delta}\left(x^2(1-\beta\delta)\right)\right]\right]\Big\} + o(\delta) \\
=& \theta V_{t+\delta}(x^1(1-\beta\delta)) + \bar{\theta} V_{t+\delta}(x^2(1-\beta\delta))
\end{aligned}$$

$$+\left(\theta(h(x^1)+\varphi)+\bar{\theta}(h(x^2)+\varphi)\right)\delta\Big\{r\left(\lambda_t^*(\bar{x})\right)+\theta\lambda_t^*(\bar{x})\left[V_{t+\delta}\left(x^1(1-\beta\delta)+\alpha(1-\beta\delta_1)\right)-V_{t+\delta}\left(x^1(1-\beta\delta)\right)\right]$$
$$+\bar{\theta}\lambda_t^*(\bar{x})\left[V_{t+\delta}\left(x^2(1-\beta\delta)+\alpha(1-\beta\delta_1)\right)-V_{t+\delta}\left(x^2(1-\beta\delta)\right)\right]\Big\}+o(\delta).$$

Since $V_{t+\delta}(x)$ is convex in x from our induction assumption, and $x^1<x^2$,

$$\begin{aligned}
&\left[\theta\left(\theta(h(x^1)+\varphi)+\bar{\theta}(h(x^2)+\varphi)\right)-\theta\left(h(x^1)+\varphi\right)\right]\left[V_{t+\delta}\left(x^1(1-\beta\delta)+\alpha(1-\beta\delta_1)\right)-V_{t+\delta}\left(x^1(1-\beta\delta)\right)\right]\\
=&\theta\bar{\theta}\left(h(x^2)-h(x^1)\right)\left[V_{t+\delta}\left(x^1(1-\beta\delta)+\alpha(1-\beta\delta_1)\right)-V_{t+\delta}\left(x^1(1-\beta\delta)\right)\right]\\
\le&\theta\bar{\theta}\left(h(x^2)-h(x^1)\right)\left[V_{t+\delta}\left(x^2(1-\beta\delta)+\alpha(1-\beta\delta_1)\right)-V_{t+\delta}\left(x^2(1-\beta\delta)\right)\right]\\
=&\left[-\bar{\theta}\left(\theta(h(x^1)+\varphi)+\bar{\theta}(h(x^2)+\varphi)\right)+\bar{\theta}\left(h(x^2)+\varphi\right)\right]\left[V_{t+\delta}\left(x^2(1-\beta\delta)+\alpha(1-\beta\delta_1)\right)-V_{t+\delta}\left(x^2(1-\beta\delta)\right)\right].
\end{aligned}$$

Therefore,

$$\begin{aligned}
V_t(\bar{x})<&\theta V_{t+\delta}(x^1(1-\beta\delta))+\bar{\theta}V_{t+\delta}(x^2(1-\beta\delta))\\
&+\theta(h(x^1)+\varphi))\delta\left\{\left[r\left(\lambda_t^*(\bar{x})\right)+\lambda_t^*(\bar{x})\left[V_{t+\delta}(x^1(1-\beta\delta)+\alpha(1-\beta\delta_1))-V_{t+\delta}(x^1(1-\beta\delta))\right]\right\}\\
&+\bar{\theta}(h(x^2)+\varphi))\delta\left\{\left[r\left(\lambda_t^*(\bar{x})\right)+\lambda_t^*(\bar{x})\left[V_{t+\delta}(x^2(1-\beta\delta)+\alpha(1-\beta\delta_1))-V_{t+\delta}(x^2(1-\beta\delta))\right]\right\}+o(\delta)\\
&\le\theta V_t(x^1)+\bar{\theta}V_t(x^2).
\end{aligned}$$

We then prove (ii) that if $h(x)$ is strictly concave in x, $V_t(x)$ is strictly concave in x by induction. Since $V_T(x)=0$, the result holds for time T. We assume $V_{t+\delta}(x)$ is concave in x for a very small $\delta>0$. We then prove that $V_t(x)$ is strictly concave in x. Let $x^1<x^2$, $\bar{x}=\theta x^1+\bar{\theta}x^2$ where $0<\theta<1$, $\bar{\theta}=1-\theta$. Denote $\lambda_i(x^i)=\arg\max_\lambda\{(h(x^i)+\varphi)r(\lambda)\delta+\lambda(h(x^i)+\varphi)\delta V_{t+\delta}((x^i(1-\beta\delta)+\alpha(1-\beta\delta_1))+(1-\lambda(h(x^i)+\varphi)\delta)[V_{t+\delta}(x^i(1-\beta\delta))]\}$ for any $i=1,2$. To make the notation simple, we use λ_i instead of $\lambda_i(x^i)$ for $i=1,2$ in the following proof. Denote $\bar{\lambda}_1=\frac{h(x^1)+\varphi}{\theta(h(x^1)+\varphi)+\bar{\theta}(h(x^2)+\varphi)}\lambda_1$, $\bar{\lambda}_2=\frac{h(x^2)+\varphi}{\theta(h(x^1)+\varphi)+\bar{\theta}(h(x^2)+\varphi)}\lambda_2$ and $\tilde{\lambda}=\theta\bar{\lambda}_1+\bar{\theta}\bar{\lambda}_2$. According to Proposition 3, $\bar{\lambda}_1<\bar{\lambda}_2$. By the HJB equation (A.1), we have

$$\begin{aligned}
&\theta V_t(x^1)+\bar{\theta}V_t(x^2)\\
=&\theta\Big[r(\lambda_1)(h(x^1)+\varphi)\delta+\lambda_1(h(x^1)+\varphi)\delta V_{t+\delta}(x^1(1-\beta\delta)+\alpha(1-\beta\delta_1))\\
&+(1-(\lambda_1(h(x^1)+\varphi))\delta)V_{t+\delta}(x^1(1-\beta\delta))\Big]\\
&+\bar{\theta}\Big[\lambda_2(h(x^2)+\varphi)p(\lambda_2)\delta+\lambda_2(h(x^1)+\varphi)\delta V_{t+\delta}(x^2(1-\beta\delta)+\alpha(1-\beta\delta_1))\\
&+(1-(\lambda_2(h(x^2)+\varphi))\delta)V_{t+\delta}(x^2(1-\beta\delta))\Big]+o(\delta)\\
=&\theta r(\lambda_1)(h(x^1)+\varphi)\delta+\bar{\theta}r(\lambda_2)(h(x^2)+\varphi)\delta\\
&+\theta\left[V_{t+\delta}\left(x^1(1-\beta\delta)\right)+\bar{\lambda}_1(h(x^1)+\varphi)\delta\left[V_{t+\delta}\left(x^1(1-\beta\delta)+\alpha(1-\beta\delta_1)\right)-V_{t+\delta}\left(x^1(1-\beta\delta)\right)\right]\right]
\end{aligned}$$

$$+\bar{\theta}\left[V_{t+\delta}\left(x^2(1-\beta\delta)\right)+\bar{\lambda}_2(h(x^2)+\varphi)\delta\left[V_{t+\delta}\left(x^2(1-\beta\delta)+\alpha(1-\beta\delta_1)\right)-V_{t+\delta}\left(x^2(1-\beta\delta)\right)\right]\right]+o(\delta).$$

Since $h(x)$ is strictly concave, $\bar{\lambda}_1(h(\bar{x})+\varphi)=\frac{h(x^1)+\varphi}{\theta(h(x^1)+\varphi)+\bar{\theta}(h(x^2)+\varphi)}\lambda_1(h(\bar{x})+\varphi)>(h(x^1)+\varphi)\lambda_1$ and $\bar{\lambda}_2(h(\bar{x})+\varphi)=\frac{h(x^2)+\varphi}{\theta(h(x^1)+\varphi)+\bar{\theta}(h(x^2)+\varphi)}\lambda_2(h(\bar{x})+\varphi)>(h(x^2)+\varphi)\lambda_2$. Thus,

$$\begin{aligned}
&\theta V_t(x^1)+\bar{\theta}V_t(x^2)\\
<&\theta r(\lambda_1)(h(x^1)+\varphi)\delta+\bar{\theta}r(\lambda_2)(h(x^2)+\varphi)\delta\\
&+\theta\left[V_{t+\delta}\left(x^1(1-\beta\delta)\right)+\bar{\lambda}_1(h(\bar{x})+\varphi)\delta\left[V_{t+\delta}\left(x^1(1-\beta\delta)+\alpha(1-\beta\delta_1)\right)-V_{t+\delta}\left(x^1(1-\beta\delta)\right)\right]\right]\\
&+\bar{\theta}\left[V_{t+\delta}\left(x^2(1-\beta\delta)\right)+\bar{\lambda}_2(h(\bar{x})+\varphi)\delta\left[V_{t+\delta}\left(x^2(1-\beta\delta)+\alpha(1-\beta\delta_1)\right)-V_{t+\delta}\left(x^2(1-\beta\delta)\right)\right]\right]+o(\delta).
\end{aligned}$$

Since $V_{t+\delta}(x)$ is concave in x from our induction assumption, and $x^1<x^2$,

$$\begin{aligned}
&\theta(\tilde{\lambda}-\bar{\lambda}_1)\left[V_{t+\delta}(x^1(1-\beta\delta)+\alpha(1-\beta\delta_1))-V_{t+\delta}\left(x^1(1-\beta\delta)\right)\right]\\
=&\theta\bar{\theta}(\bar{\lambda}_2-\bar{\lambda}_1)\left[V_{t+\delta}\left(x^1(1-\beta\delta)+\alpha(1-\beta\delta_1)\right)-V_{t+\delta}\left(x^1(1-\beta\delta)\right)\right]\\
\geq&\theta\bar{\theta}(\bar{\lambda}_2-\bar{\lambda}_1)\left[V_{t+\delta}\left(x^2(1-\beta\delta)+\alpha(1-\beta\delta_1)\right)-V_{t+\delta}\left(x^2(1-\beta\delta)\right)\right]\\
=&\bar{\theta}(\bar{\lambda}_2-\tilde{\lambda})\left[V_{t+\delta}\left(x^2(1-\beta\delta)+\alpha(1-\beta\delta_1)\right)-V_{t+\delta}\left(x^2(1-\beta\delta)\right)\right].
\end{aligned}$$

Therefore,

$$\begin{aligned}
&\theta V_t(x^1)+\bar{\theta}V_t(x^2)\\
<&\theta r(\lambda_1)(h(x^1)+\varphi)\delta+\bar{\theta}r(\lambda_2)(h(x^2)+\varphi)\delta+\theta V_{t+\delta}\left(x^1(1-\beta\delta)\right)+\bar{\theta}V_{t+\delta}\left(x^2(1-\beta\delta)\right)\\
&+\theta\tilde{\lambda}(h(\bar{x})+\varphi)\delta\left[V_{t+\delta}\left(x^1(1-\beta\delta)+\alpha(1-\beta\delta_1)\right)-V_{t+\delta}\left(x^1(1-\beta\delta)\right)\right]\\
&+\bar{\theta}\tilde{\lambda}(h(\bar{x})+\varphi)\delta\left[V_{t+\delta}(x^2(1-\beta\delta)+\alpha(1-\beta\delta_1))-V_{t+\delta}(x^2(1-\beta\delta))\right]+o(\delta).
\end{aligned}$$

Let $\theta_h=\frac{\theta(h(x^1)+\varphi)}{\theta(h(x^1)+\varphi)+\bar{\theta}(h(x^2)+\varphi)}$, and $\bar{\theta}_h=1-\theta_h$. Since $h(x^1)<h(x^2)$, $0<\theta_h<1$. Since $r(\lambda)$ and $h(x)$ are both concave,

$$\begin{aligned}
&\theta r(\lambda_1)(h(x^1)+\varphi)+\bar{\theta}r(\lambda_2)(h(x^2)+\varphi)\\
=&\left[\theta(h(x^1)+\varphi)+\bar{\theta}(h(x^2)+\varphi)\right]\left[\theta_h r(\lambda_1)+\bar{\theta}_h r(\lambda_2)\right]\\
\leq&\left[\theta(h(x^1)+\varphi)+\bar{\theta}(h(x^2)+\varphi)\right]r(\theta_h\lambda_1+\bar{\theta}_h\lambda_2)\\
<&(h(\bar{x})+\varphi)r(\tilde{\lambda}).
\end{aligned}$$

Thus,

$$\theta V_t(x^1)+\bar{\theta}V_t(x^2)$$

$$
\begin{aligned}
<&(h(\bar{x})+\varphi)r(\tilde{\lambda})\delta+\theta V_{t+\delta}\left(x^1(1-\beta\delta)\right)+\bar{\theta}V_{t+\delta}\left(x^2(1-\beta\delta)\right)\\
&+\theta\tilde{\lambda}(h(\bar{x})+\varphi)\delta\left[V_{t+\delta}\left(x^1(1-\beta\delta)+\alpha(1-\beta\delta_1)\right)-V_{t+\delta}\left(x^1(1-\beta\delta)\right)\right]\\
&+\bar{\theta}\tilde{\lambda}(h(\bar{x})+\varphi)\delta\left[V_{t+\delta}(x^2(1-\beta\delta)+\alpha(1-\beta\delta_1))-V_{t+\delta}(x^2(1-\beta\delta))\right]+o(\delta)\\
=&(h(\bar{x})+\varphi)r(\tilde{\lambda})\delta\\
&+\tilde{\lambda}(h(\bar{x})+\varphi)\delta\left[\theta V_{t+\delta}\left(x^1(1-\beta\delta)+\alpha(1-\beta\delta_1)\right)+\bar{\theta}V_{t+\delta}\left(x^2(1-\beta\delta)+\alpha(1-\beta\delta_1)\right)\right]\\
&+\left[1-\tilde{\lambda}(h(\bar{x})+\varphi)\delta\right]\left[\theta V_{t+\delta}\left(x^1(1-\beta\delta)\right)+\bar{\theta}V_{t+\delta}\left(x^2(1-\beta\delta)\right)\right]+o(\delta)\\
<&(h(\bar{x})+\varphi)r(\tilde{\lambda})\delta+\tilde{\lambda}(h(\bar{x})+\varphi)\delta V_{t+\delta}(\bar{x}(1-\beta\delta)+\alpha(1-\beta\delta_1))+(1-\tilde{\lambda}(h(\bar{x})+\varphi)\delta)V_{t+\delta}(\bar{x}(1-\beta\delta))+o(\delta)\\
\leq&V_t(\bar{x}).
\end{aligned}
$$

This completes the proof.

We next prove (iii) that the value function $V_t(x)$ is linear in x by induction. Since the last time valuation $V_T(x)=0$, which is linear in x. We assume $V_{t+\delta}(x)$ is linear in x, we then prove $V_t(x)$ is linear in x as follows. From the HJB equation (A.1) and $h(x)=h_1x+h_0$, we have

$$
\begin{aligned}
V_t(x)=&\max_{\lambda}\{(h_1x+h_0+\varphi)r(\lambda)\delta+\lambda(h_1x+h_0+\varphi)\delta V_{t+\delta}(x-\beta\delta x+\alpha(1-\beta\delta_1))\\
&+(1-\lambda(h_1x+h_0+\varphi)\delta)[V_{t+\delta}(x-\beta\delta x)]\}+o(\delta)\\
=&V_{t+\delta}(x-\beta\delta x)+(h_1x+h_0+\varphi)\delta\max_{\lambda}\{r(\lambda)+\lambda(V_{t+\delta}(x-\beta\delta x+\alpha(1-\beta\delta_1))-V_{t+\delta}(x-\beta\delta x))\}+o(\delta).
\end{aligned}
$$

Since $V_{t+\delta}(x)$ is linear in x, $V_{t+\delta}(x-\beta\delta x)$ is linear in x, $V_{t+\delta}(x-\beta\delta x+\alpha(1-\beta\delta_1))-V_{t+\delta}(x-\beta\delta x)$ is independent of x. Therefore, $\max_\lambda\{r(\lambda)+\lambda(V_{t+\delta}(x-\beta\delta x+\alpha(1-\beta\delta_1))-V_{t+\delta}(x-\beta\delta x))\}$ is independent of x, in which, we have $\lambda_t^*(x)$ is independent of x. Thus $(h_1x+h_0+\varphi)\delta\max_\lambda\{r(\lambda)+\lambda(V_{t+\delta}(x-\beta\delta x+\alpha(1-\beta\delta_1))-V_{t+\delta}(x-\beta\delta x))\}$ is linear in x. $V_t(x)$ is the sum of two linear functions, which is linear in x. □

Proof of Corollary 1. From Theorem 2, the value function must be linear if $h(x)$ is linear.

Let $V_t(x)=\mu_tx+\omega_t$, where μ_t and ω_t are independent of x. Since $V_T(x)=0$ for any x, $\mu_T=0$ and $\omega_T=0$. From Equation (3),

$$
-\frac{\partial}{\partial t}(\mu_tx+\omega_t)+x\beta\frac{\partial}{\partial x}(\mu_tx+\omega_t)=(h_1x+h_0+\varphi)\max_{\lambda}\left\{r(\lambda)+\lambda\left[\mu_t(x+\alpha)+\omega_t-(\mu_tx+\omega_t)\right]\right\}.
$$

Simplifying the above equation , we have

$$
\left(\beta\mu_t-\frac{d\mu_t}{dt}\right)x+\frac{d\omega_t}{dt}=(h_1x+h_0+\varphi)\max\{r(\lambda)+\lambda\alpha\mu_t\}.
$$

Note that the above equation holds for any x. By comparing the coefficient of x and that of the constant, we have

$$
-\frac{d\mu_t}{dt}+\beta\mu_t=h_1\max\{r(\lambda)+\lambda\alpha\mu_t\},-\frac{d}{dt}\omega_t=(h_0+\varphi)\max\{r(\lambda)+\lambda\alpha\mu_t\}.
$$

Therefore,

$$\frac{d\mu_t}{dt} = \beta\mu_t - h_1 \max\{r(\lambda) + \lambda\alpha\mu_t\} \quad \text{and} \quad \frac{d\mu_t}{dt} = \beta\mu_t + \frac{h_1}{h_0 + \varphi}\frac{d\omega_t}{dt}. \quad \square$$

Proof of Theorem 3. We first prove that $E[V_s(x_s+\alpha) - V_s(x_s)|\mathcal{F}_t] \le V_t(x_t+\alpha) - V_t(x_t)$ when $\beta = 0$. To prove the above inequality, we only need to prove $E[V_{t+\delta}(x_{t+\delta}+\alpha) - V_{t+\delta}(x_{t+\delta})|\mathcal{F}_t] \le V_t(x_t+\alpha) - V_t(x_t)$ for a very small $\delta > 0$. When $x_t = x$, $x_{t+\delta} = x + \alpha - \beta x\delta$ w.p. $\Lambda_t^*(x)\delta$ and $x_{t+\delta} = x - \beta x\delta$ w.p. $1 - \Lambda_t^*(x)\delta$. Therefore,

$$\begin{aligned}
&E[V_{t+\delta}(x_{t+\delta}+\alpha) - V_{t+\delta}(x_{t+\delta})|\mathcal{F}_t] \\
=&V_{t+\delta}(x+\alpha-\beta x\delta)(1-\Lambda_t^*(x)\delta) + V_{t+\delta}(x+2\alpha-\beta(x+\alpha)\delta)\Lambda_t^*(x)\delta \\
&- V_{t+\delta}(x-\beta x\delta)(1-\Lambda_t^*(x)\delta) - V_{t+\delta}(x+\alpha-\beta(x+\alpha)\delta)\Lambda_t^*(x)\delta + o(\delta).
\end{aligned}$$

We can reexpress the value function in $t+\delta$ by the Taylor expression:

$$\begin{aligned}
&E[V_{t+\delta}(x_{t+\delta}+\alpha) - V_{t+\delta}(x_{t+\delta})|\mathcal{F}_t] \\
=&\left[V_t(x+\alpha) + \frac{\partial}{\partial t}V_t(x+\alpha)\delta - \beta x\delta\frac{\partial}{\partial x}V_t(x+\alpha)\right](1-\Lambda_t^*(x)\delta) \\
&+ \left[V_t(x+2\alpha) + \frac{\partial}{\partial t}V_t(x+2\alpha)\delta - \beta(x+\alpha)\delta\frac{\partial}{\partial x}V_t(x+2\alpha)\right]\Lambda_t^*(x)\delta \\
&- \left[V_t(x) + \frac{\partial}{\partial t}V_t(x)\delta - \beta x\delta\frac{\partial}{\partial x}V_t(x)\right](1-\Lambda_t^*(x)\delta) \\
&- \left[V_t(x+\alpha) + \frac{\partial}{\partial t}V_t(x+\alpha)\delta - \beta(x+\alpha)\delta\frac{\partial}{\partial x}V_t(x+\alpha)\right]\Lambda_t^*(x)\delta + o(\delta) \\
=&V_t(x+\alpha)(1-\Lambda_t^*(x)\delta) + \frac{\partial}{\partial t}V_t(x+\alpha)\delta - \beta x\delta\frac{\partial}{\partial x}V_t(x+\alpha) + V_t(x+2\alpha)\Lambda_t^*(x)\delta \\
&- V_t(x)(1-\Lambda_t^*(x)\delta) - \frac{\partial}{\partial t}V_t(x)\delta + \beta x\delta\frac{\partial}{\partial x}V_t(x) - V_t(x+\alpha)\Lambda_t^*(x)\delta + o(\delta) \\
=&V_t(x+\alpha) - V_t(x) + \alpha\beta\delta\frac{\partial}{\partial x}V_t(x+\alpha) \\
&+ \delta\left[(h(x)+\varphi)r(\lambda_t^*(x)) + \frac{\partial}{\partial t}V_t(x+\alpha) - \beta(x+\alpha)\frac{\partial}{\partial x}V_t(x+\alpha) + (V_t(x+2\alpha) - V_t(x+\alpha))\Lambda_t^*(x)\right] \\
&- \delta\left[(h(x)+\varphi)r(\lambda_t^*(x)) + \frac{\partial}{\partial t}V_t(x) - \beta(x)\frac{\partial}{\partial x}V_t(x) + (V_t(x+\alpha) - V_t(x))\Lambda_t^*(x)\right] + o(\delta) \\
=&V_t(x+\alpha) - V_t(x) + \alpha\beta\delta\frac{\partial}{\partial x}V_t(x+\alpha) \\
&+ \delta\left[(h(x)+\varphi)r(\lambda_t^*(x)) + \frac{\partial}{\partial t}V_t(x+\alpha) - \beta(x+\alpha)\frac{\partial}{\partial x}V_t(x+\alpha) + (V_t(x+2\alpha) - V_t(x+\alpha))\Lambda_t^*(x)\right].
\end{aligned}$$

Since $h(x+\alpha) \ge h(x)$,

$$E[V_{t+\delta}(x_{t+\delta}+\alpha) - V_{t+\delta}(x_{t+\delta})|\mathcal{F}_t]$$

$$
\begin{aligned}
&\le V_t(x+\alpha) - V_t(x) + \alpha\beta\delta\frac{\partial}{\partial x}V_t(x+\alpha) \\
&+ \delta\max_{\lambda}\Big\{(h(x+\alpha)+\varphi)r(\lambda) + \frac{\partial}{\partial t}V_t(x+\alpha) - \beta(x+\alpha)\frac{\partial}{\partial x}V_t(x+\alpha) \\
&+ (V_t(x+2\alpha) - V_t(x+\alpha))\lambda(h(x+\alpha)+\varphi)\Big\}.
\end{aligned}
$$

From the HJB equation (3), $-\frac{\partial}{\partial t}V_t(x+\alpha) + \beta(x+\alpha)\frac{\partial}{\partial x}V_t(x+\alpha) = \max_\lambda\Big\{(h(x+\alpha)+\varphi)r(\lambda) + (V_t(x+2\alpha) - V_t(x+\alpha))\lambda(h(x+\alpha)+\varphi)\Big\}$. Finally, since $\beta = 0$, $E[V_s(x_s+\alpha) - V_s(x_s)|\mathcal{F}_t] \le V_t(x_t+\alpha) - V_t(x_t)$.

For any $x \ge 0$ and $t < T$, the price can be solved by

$$
\max_{p\ge 0}\{\lambda(p)(p - c + V_t(x+\alpha) - V_t(x))\}. \tag{A.2}
$$

Therefore, the marginal profit function is $\lambda(p) + \lambda'(p)(p - c + V_t(x+\alpha) - V_t(x)) = \lambda'(p)(H(p) + V_t(x+\alpha) - V_t(x))$, where $H(p) = p - c - 1/g(p)$ and $g(p) = -\lambda'(p)/\lambda(p)$. $1/g(p)$ is the Mills ratio.

Following the same steps of Lemma 1 in Pang et al. (2014), we can show that $H(p)$ is strictly increasing in p, for $p \ge 0$. Following the same steps of the proof of Theorem 1 in Pang et al. (2014). We have that the convexity of $1/g(p)$ implies the convexity and increasing of $H^{-1}(x)$ in x.

Then for any $0 \le t < s \le T$, using Jensen's Inequality,

$$
E[p_s^*(x_s)|\mathcal{F}_t] = E\left[H^{-1}(-(V_s(x_s+\alpha) - V_s(x_s)))|\mathcal{F}_t\right] \ge H^{-1}\left(E[-(V_s(x_s+\alpha) - V_s(x_s)))|\mathcal{F}_t]\right).
$$

Note that $H^{-1}(x)$ increases in x. Also $E[V_s(x_s+\alpha) - V_s(x_s)|\mathcal{F}_t] \le V_t(x_t+\alpha) - V_t(x_t)$ from the first part of the proof,

$$
E[p_s^*(x_s)|\mathcal{F}_t] \ge H^{-1}(-(V_t(x_t+\alpha) - V_t(x_t))) = p_t^*(x_t). \quad \square
$$

Proof of Proposition 4. For Markov policies, $\lambda_{t+\delta}^u$ is determined by the filtration $\mathcal{F}_t$.

With probability $n\lambda_t^u\left[\varphi + h\left(x_t^u/n\right)\right]\delta$, $x_{t+\delta}^u = x_t^u + \alpha - \beta x_t^u\delta$. With probability $1 - n\lambda_t^u\left[\varphi + h\left(x_t^u/n\right)\right]\delta$, $x_{t+\delta}^u = x_t^u - \beta x_t^u\delta$. Thus,

$$
\mathbb{E}(x_{t+\delta}^u|F_t) = (1 - \Lambda_t^u\delta)(x_t^u - \beta x_t^u\delta) + \Lambda_t^u\delta(x_t^u + \alpha - \beta x_t^u\delta) = x_t^u + (\alpha\Lambda_t^u - \beta x_t^u)\,\delta.
$$

Take expectation on both sides, and let $\delta \to 0$,

$$
\frac{d}{dt}\mathbb{E}[x_t^u] = \alpha n\mathbb{E}[\lambda_t^u]\left[\varphi + h\left(\frac{x_t^u}{n}\right)\right] - \beta\mathbb{E}[x_t^u],
$$

with boundary condition $\mathbb{E}x_0^u = 0$. Since $h(\cdot)$ is Lipschitz continuous, $h\left(x_t^u/n\right) \le \kappa x_t^u/n$, where κ is a constant. Thus,

$$
\frac{d}{dt}\mathbb{E}[x_t^u] \le \alpha\varphi\bar{\lambda}n + (\alpha\bar{\lambda}\kappa - \beta)\mathbb{E}[x_t^u].
$$

$\mathbb{E}[x_t^u] = O(n)$ follows immediately using Gronwall's Inequality. $\square$

To prove Proposition 5, we prove two auxiliary results. First in Lemma 5, we provide bounds of the variance of x_t^u for a subset of policies where the variance of λ_t^u is "small." With a slight abuse of notation, we omit the scale n when there is no confusion.

LEMMA 5. *For any Markov policy u such that* $\mathrm{Var}[\lambda_t^u] = O(\mathrm{Var}[x_t^u]/n^2)$ *for any* $t \geq 0$, $\mathrm{Var}[x_t^u] = O(n)$ *and* $\mathrm{Var}[\lambda_t^u] = O(1/n)$.

Proof of Lemma 5. At time t with the realized x_t^u,

$$\mathbb{E}(x_{t+\delta}^u|\mathcal{F}_t) = x_t^u + (\alpha\Lambda_t^u - \beta x_t^u)\,\delta = x_t^u + \left[\alpha n\lambda_t^u\left(\varphi + h\left(\frac{x_t^u}{n}\right)\right) - \beta x_t^u\right]\delta,$$

from the proof of Proposition 4. Also

$$\begin{aligned}
&\mathrm{Var}\left[x_{t+\delta}^u|\mathcal{F}_t\right] = \mathbb{E}\left[\left(x_{t+\delta}^u\right)^2|\mathcal{F}_t\right] - \left(\mathbb{E}\left[x_{t+\delta}^u|\mathcal{F}_t\right]\right)^2 \\
&= (1-\Lambda_t^u\delta)(x_t^u - \beta x_t^u\delta)^2 + \Lambda_t^u\delta(x_t^u + \alpha - \beta x_t^u\delta)^2 - (x_t^u - \beta x_t^u\delta + \alpha\Lambda_t^u\delta)^2 \\
&= \Lambda_t^u\delta\left[\alpha^2 + 2\alpha(x_t^u - \beta x_t^u\delta)\right] - 2\alpha\Lambda_t^u\delta(x_t^u - \beta x_t^u\delta) + o(\delta) \\
&= \alpha^2\Lambda_t^u\delta + o(\delta).
\end{aligned}$$

Therefore,

$$\begin{aligned}
\mathrm{Var}[x_{t+\delta}^u] &= \mathbb{E}\left[\mathrm{Var}[x_{t+\delta}^u|\mathcal{F}_t]\right] + \mathrm{Var}\left[\mathbb{E}\left[x_{t+\delta}^u\middle|\mathcal{F}_t\right]\right] \\
&= \mathrm{Var}[x_t^u] + 2\delta\mathrm{Cov}\left(\alpha n\lambda_t^u\left[\varphi + h\left(\frac{x_t^u}{n}\right)\right] - \beta x_t^u, x_t^u\right) + \alpha^2\mathbb{E}\Lambda_t^u\delta + o(\delta).
\end{aligned}$$

Rearrange the terms and let $\delta \to 0$,

$$\begin{aligned}
\frac{d}{dt}\mathrm{Var}[x_t^u] &= 2\alpha\varphi n\mathrm{Cov}(\lambda_t^u, x_t^u) + 2n\mathrm{Cov}\left(\lambda_t^u h\left(\frac{x_t^u}{n}\right), x_t^u\right) - 2\beta\mathrm{Var}(x_t^u) + O(n) \\
&\leq 2\alpha\varphi n\sqrt{\mathrm{Var}[\lambda_t^u]\mathrm{Var}[x_t^u]} + 2n\sqrt{\mathrm{Var}\left[\lambda_t^u h\left(\frac{x_t^u}{n}\right)\right]\mathrm{Var}[x_t^u]} - 2\beta\mathrm{Var}[x_t^u] + O(n).
\end{aligned}$$

$\mathrm{Var}\left[\lambda_t^u h\left(\frac{x_t^u}{n}\right)\right] \leq 2\bar{\lambda}^2\mathrm{Var}\left[h\left(\frac{x_t^u}{n}\right)\right] + 2\left(\mathbb{E}h\left[\frac{x_t^u}{n}\right]\right)^2\mathrm{Var}[\lambda_t^u]$. Since $h(\cdot)$ is Lipschitz continuous, $\mathrm{Var}\left[h\left(x_t^u/n\right)\right] = O\left(\mathrm{Var}[x_t^u]/n^2\right)$ and $\mathbb{E}h\left(x_t^u/n\right) = O\left(\mathbb{E}[x_t^u]/n\right) = O(1)$. Therefore, $\mathrm{Var}\left[\lambda_t^u h\left(x_t^u/n\right)\right] = O\left(\mathrm{Var}[(x_t^u]/n^2\right)$.

Thus there exists constants k_1 and k_2, such that $\frac{d}{dt}\mathrm{Var}[x_t^u] \leq k_1\mathrm{Var}[x_t^u] + k_2 n$ when n is sufficiently large. Using Gronwall's Inequality, $\mathrm{Var}[x_t^u] = O(n)$. □

Next we prove a lemma that bounds $\frac{\partial^k}{\partial x^k}V_t^{(n)}(x)$ in Lemma 6.

LEMMA 6. *For $k \geq 1$ and $x \leq \rho n$, where ρ is a constant,* $\frac{\partial^k}{\partial x^k}V_t^{(n)}(x) = O\left(1/n^{k-1}\right)$.

Proof of Lemma 6. First the claim holds for $t=T$, as $V_t^{(n)}(x)=0$ by definition. We show that if the claim holds for any $t\le T$, it must also hold for $t-\delta$. To do that, we just need to show that $\frac{\partial}{\partial t}\frac{\partial^k V_t^{(n)}(x)}{\partial x^k}=O\left(\frac{1}{n^{k-1}}\right)$ for any k.

Let $R_t^{(n)}(x)=r(\lambda_t^{(n)}(x))+\lambda_t^{(n)}(x)\left(V_t^{(n)}(x+\alpha)-V_t^{(n)}(x)\right)$. Then $-\frac{\partial}{\partial t}V_t^{(n)}(x)+\beta x\frac{\partial}{\partial x}V_t^{(n)}(x)=n\left[\varphi+h\left(\frac{x}{n}\right)\right]R_t^{(n)}(x)$. We can thus write $\frac{\partial}{\partial t}\frac{\partial^k V_t^{(n)}(x)}{\partial x^k}$ as

$$\frac{\partial}{\partial t}\frac{\partial^k V_t^{(n)}(x)}{\partial x^k}=\beta\frac{\partial^k}{\partial x^k}\left(x\frac{\partial}{\partial x}V_t^{(n)}(x)\right)-\sum_{j=0}^{k}n\frac{\partial^{k-j}}{\partial x^{k-j}}R_t^{(n)}(x)\frac{\partial^j}{\partial x^j}\left[\varphi+h\left(\frac{x}{n}\right)\right].$$

Since $x=O(n)$, $\frac{\partial^k}{\partial x^k}\left(x\frac{\partial}{\partial x}V_t^{(n)}(x)\right)=O(\frac{1}{n^{k-1}})$. Also $\frac{\partial^j}{\partial x^j}h\left(\frac{x}{n}\right)=O\left(\frac{1}{n^j}\right)$. Now for $\frac{\partial^m R_t^{(n)}(x)}{\partial x^m}$:

$$\frac{\partial^m R_t^{(n)}(x)}{\partial x^m}=\frac{\partial^m}{\partial x^m}r(\lambda_t^{(n)}(x))+\sum_{j=0}^{m}\frac{\partial^{m-j}}{\partial x^{m-j}}\lambda_t^{(n)}(x)\frac{\partial^j}{\partial x^j}\left[V_t^{(n)}(x+\alpha)-V_t^{(n)}(x)\right].$$

According to our induction assumption, $\frac{\partial^m}{\partial x^m}\left[V_t^{(n)}(x+\alpha)-V_t^{(n)}(x)\right]=\alpha\frac{\partial^{m+1}}{\partial x^{m+1}}V_t^{(n)}(x)\Big|_{x=x+\theta\alpha}=O\left(\frac{1}{n^m}\right)$, where $\theta\in[0,1]$. Since $\lambda_t^{(n)}(x)$ is the solution of $r'(\lambda)=-\left[V_t^{(n)}(x+\alpha)-V_t^{(n)}(x)\right]$, $\frac{\partial^m}{\partial x^m}\lambda_t^{(n)}(x)=O\left(\frac{1}{n^m}\right)$, which implies that $\frac{\partial^m}{\partial x^m}r(\lambda_t^{(n)}(x))=O(\frac{1}{n^m})$. Therefore,

$$\frac{\partial^m R_t^{(n)}(x)}{\partial x^m}=\frac{\partial^m}{\partial x^m}r(\lambda_t^{(n)}(x))+\sum_{j=0}^{m}\frac{\partial^{m-j}}{\partial x^{m-j}}\lambda_t^{(n)}(x)\frac{\partial^j}{\partial x^j}\left[V_t^{(n)}(x+\alpha)-V_t^{(n)}(x)\right]=O\left(\frac{1}{n^m}\right).$$

Thus

$$\frac{\partial}{\partial t}\frac{\partial^k V_t^{(n)}(x)}{\partial x^k}=\beta\frac{\partial^k}{\partial x^k}\left(x\frac{\partial}{\partial x}V_t^{(n)}(x)\right)-\sum_{j=0}^{k}n\frac{\partial^{k-j}}{\partial x^{k-j}}R_t^{(n)}(x)\frac{\partial^j}{\partial x^j}\left[\varphi+h\left(\frac{x}{n}\right)\right]=O\left(\frac{1}{n^{k-1}}\right).$$

This completes the proof. □

Proof of Proposition 5.

$$\begin{aligned}
\mathrm{Var}[\lambda_t^*] &= \mathrm{Var}[\lambda_t^*(x_{t-}^*)]-\lambda_t^*(\mathbb{E}[x_t^*]))=\mathrm{Var}\left[\int_{\mathbb{E}[x_t^*]}^{x_{t-}^*}\frac{\partial}{\partial x}\lambda_t^{(n)}(x)dx\right]\le\mathbb{E}\left[\int_{\mathbb{E}[x_t^*]}^{x_{t-}^*}\frac{\partial}{\partial x}\lambda_t^{(n)}(x)dx\right]^2\\
&=\mathbb{E}\left\{\left[\int_{\mathbb{E}[x_t^*]}^{x_{t-}^*}\frac{\partial}{\partial x}\lambda_t^{(n)}(x)dx\right]^2\middle| x_t^*\ge\mathbb{E}[x_t^*]+n\right\}\mathbb{P}\left(x_t^*\ge\mathbb{E}[x_t^*]+n\right)+\\
&\quad\mathbb{E}\left\{\left[\int_{\mathbb{E}[x_t^*]}^{x_{t-}^*}\frac{\partial}{\partial x}\lambda_t^{(n)}(x)dx\right]^2\middle| x_t^*<\mathbb{E}[x_t^*]+n\right\}\mathbb{P}\left(x_t^*<\mathbb{E}[x_t^*]+n\right)\\
&\le\bar{\lambda}^2\mathbb{P}\left(x_t^*\ge\mathbb{E}[x_t^*]+n\right)+\mathbb{E}\left\{\left[\int_{\mathbb{E}[x_t^*]}^{x_{t-}^*}\frac{\partial}{\partial x}\lambda_t^{(n)}(x)dx\right]^2\middle| x_t^*<\mathbb{E}[x_t^*]+n\right\}.
\end{aligned}$$

By applying Lemma 6 at $k=2$, $\frac{\partial}{\partial x}\lambda_t^{(n)}(x)=O(1/n)$ for any $x<\mathbb{E}[x_t^*]+n$, since $\mathbb{E}[x_t^*]=O(n)$

according to Lemma 4. Using Chebyshev's inequality, $\mathbb{P}\left(x_t^* \geq \mathbb{E}[x_t^*] + n\right) \leq \mathrm{Var}[x_t^*]/n^2$. Therefore there exists a constant κ_1 independent of n such that

$$\mathrm{Var}[\lambda_t^*] \leq \frac{\bar{\lambda}^2}{n^2}\mathrm{Var}[x_t^*] + \frac{\kappa_1}{n^2}\mathbb{E}\left[x_t^* - \mathbb{E}[x_t^*]\right]^2 = \frac{\bar{\lambda}^2 + \kappa_1}{n^2}\mathrm{Var}[x_t^*].$$

Since $\mathrm{Var}[\lambda_0^*] = 0$, $\mathrm{Var}[\lambda_t^*] = O(\mathrm{Var}[x_t^*]/n^2)$. The result then follows immediately from Lemma 5.

□

Before we prove Theorem 4, we establish the following lemma that bounds the expectation of a function of a series of bounded random variables, when the variance vanishes and the function is Lipschitz continuous.

LEMMA 7. *Consider a series of random variables $X_n > 0$ that have bounded expectations, i.e. $\mathbb{E}X_n \leq B$ where B is a constant independent of n. If $\mathrm{Var}[X_n] \to 0$ as $n \to \infty$, $\mathbb{E}h(X_n) - h(\mathbb{E}X_n) = O(\mathrm{Var}[X_n])$ for any increasing Lipschitz continuous function $h(\cdot)$.*

Proof of Lemma 7. Without loss of generality, assume we assume that $h(0) = 0$. Since $h(x)$ is an increasing Lipschitz continuous function, we assume that $h'(x) \in [0, L]$, where L is a positive constant independent of x. We obtain the lower bound first. Using Taylor's expansion,

$$\begin{aligned}
&\mathbb{E}h(X_n) > \mathbb{E}\left[h(X_n)|X \leq \mathbb{E}X + 1\right]\mathbb{P}\left(X_n \leq \mathbb{E}X_n + 1\right) \\
&= \mathbb{E}\left[h(\mathbb{E}X_n) + h'((1-\theta)X_n + \theta\mathbb{E}X_n)(X_n - \mathbb{E}X_n)|X_n \leq \mathbb{E}X_n + 1\right]\mathbb{P}\left(X_n \leq \mathbb{E}X_n + 1\right),
\end{aligned}$$

where $\theta \in [0, 1]$. Since $\mathbb{E}\left[X_n - \mathbb{E}X_n|X \leq \mathbb{E}X_n + 1\right] \leq 0$ and $h'(\cdot) \leq L$,

$$\mathbb{E}h(X_n) \geq h(\mathbb{E}X_n) - h(\mathbb{E}X_n)\mathbb{P}\left(X_n > \mathbb{E}X_n + 1\right) + L\mathbb{E}\left[X_n - \mathbb{E}X_n|X_n \leq \mathbb{E}X_n + 1\right].$$

Using Theorem 2.1 in Mallows and Richter 1969, we have

$$\mathbb{E}\left[X_n - \mathbb{E}X_n|X_n \leq \mathbb{E}X_n + 1\right] \geq -\sqrt{\mathrm{Var}[X_n]\frac{\mathbb{P}\left(X_n > \mathbb{E}X_n + 1\right)}{1 - \mathbb{P}\left(X_n > \mathbb{E}X_n + 1\right)}}.$$

Therefore

$$\mathbb{E}h(X_n) \geq h(\mathbb{E}X_n) - h(\mathbb{E}X_n)\mathbb{P}\left(X_n > \mathbb{E}X_n + 1\right) - L\sqrt{\mathrm{Var}[X_n]\frac{\mathbb{P}\left(X_n > \mathbb{E}X_n + 1\right)}{1 - \mathbb{P}\left(X_n > \mathbb{E}X_n + 1\right)}}.$$

Using Chebyshev's Inequality, $\mathbb{P}\left(X_n > \mathbb{E}X_n + 1\right) \leq \mathrm{Var}[X_n]$. Also since and $\mathbb{E}X_n \leq B$ and $\mathrm{Var}[X_n] \to 0$,

$$\mathbb{E}h(X_n) \geq h(\mathbb{E}X_n) - h(B)\mathrm{Var}[X_n] - L\sqrt{\frac{\mathrm{Var}^2[X_n]}{1 - \mathrm{Var}[X_n]}} \geq h(\mathbb{E}X_n) - \left(\frac{L}{2} + h(B)\right)\mathrm{Var}[X_n],$$

when n is sufficiently large. Next we obtain the upper bound. Since $h(0)=0$ and $h'(x)\le L$, $h(x)\le Lx$ for any x. Therefore

$$\begin{aligned}
&\mathbb{E}h(X_n) = \mathbb{E}\left[h(X_n)|X_n\le \mathbb{E}X_n+1\right]\mathbb{P}\left(X_n\le \mathbb{E}X_n+1\right)+\mathbb{E}\left[h(X_n)|X_n> \mathbb{E}X_n+1\right]\mathbb{P}\left(X_n> \mathbb{E}X_n+1\right)\\
&\le \mathbb{E}\left[h(X_n)|X_n\le \mathbb{E}X_n+1\right]\mathbb{P}\left(X_n\le \mathbb{E}X_n+1\right)+L\mathbb{E}\left[X_n|X_n> \mathbb{E}X_n+1\right]\mathbb{P}\left(X_n> \mathbb{E}X_n+1\right)\\
&= \mathbb{E}\left[h(\mathbb{E}X_n)+h'\left(\mathbb{E}X_n\right)\left(X_n-\mathbb{E}X_n\right)+h''((1-\theta)X_n+\theta\mathbb{E}X_n)(X_n-\mathbb{E}X_n)^2\big|X_n\le \mathbb{E}X_n+1\right]\mathbb{P}\left(X_n\le \mathbb{E}X_n+1\right)\\
&\quad +L\mathbb{E}\left[X_n|X_n> \mathbb{E}X_n+1\right]\mathbb{P}\left(X_n> \mathbb{E}X_n+1\right)
\end{aligned}$$

Since $\mathbb{E}\left[X_n-\mathbb{E}X_n|X_n\le \mathbb{E}X_n+1\right]\le 0$ and $h'(x)\ge 0$,

$$\begin{aligned}
&\mathbb{E}h(X_n)\le \mathbb{E}\left[h(\mathbb{E}X_n)+h''((1-\theta)X_n+\theta\mathbb{E}X_n)(X_n-\mathbb{E}X_n)^2\big|X_n\le \mathbb{E}X_n+1\right]\mathbb{P}\left(X_n\le \mathbb{E}X_n+1\right)\\
&+L\mathbb{E}\left[X_n|X_n> \mathbb{E}X_n+1\right]\mathbb{P}\left(X_n> \mathbb{E}X_n+1\right).
\end{aligned}$$

Since $\mathbb{E}X_n$ is bounded, there exists a constant b such that $|h''(x)|\le b$ for any $x\le \mathbb{E}X_n+1$. Therefore

$$\begin{aligned}
\mathbb{E}h(X_n) \le\ & h(\mathbb{E}X_n)+b\mathbb{E}\left[(X_n-\mathbb{E}X_n)^2\big|X_n\le \mathbb{E}X_n+1\right]\mathbb{P}\left(X_n\le \mathbb{E}X_n+1\right)\\
&+L\mathbb{E}\left[X_n|X_n> \mathbb{E}X_n+1\right]\mathbb{P}\left(X_n> \mathbb{E}X_n+1\right)\\
\le\ & h(\mathbb{E}X_n)+b\mathrm{Var}[X]+L\mathbb{E}\left[X_n|X_n> \mathbb{E}X_n+1\right]\mathbb{P}\left(X_n> \mathbb{E}X_n+1\right).
\end{aligned}$$

Again using Theorem 2.1 in Mallows and Richter 1969, we have

$$\mathbb{E}\left[X_n-\mathbb{E}X_n|X_n> \mathbb{E}X_n+1\right]\le\sqrt{\mathrm{Var}[X_n]\frac{1-\mathbb{P}\left(X_n> \mathbb{E}X_n+1\right)}{\mathbb{P}\left(X_n> \mathbb{E}X_n+1\right)}}\le\sqrt{\frac{\mathrm{Var}[X_n]}{\mathbb{P}\left(X> \mathbb{E}X+1\right)}}.$$

Therefore,

$$\begin{aligned}
&\mathbb{E}h(X_n)\le h(\mathbb{E}X_n)+b\mathrm{Var}[X]+L\left(\mathbb{E}X_n+\sqrt{\frac{\mathrm{Var}[X_n]}{\mathbb{P}\left(X_n> \mathbb{E}X_n+1\right)}}\right)\mathbb{P}\left(X_n> \mathbb{E}X_n+1\right)\\
&\le h(\mathbb{E}X_n)+b\mathrm{Var}[X_n]+L\left(\mathbb{E}X_n\cdot\mathbb{P}\left(X_n> \mathbb{E}X_n+1\right)+\sqrt{\mathrm{Var}[X_n]\mathbb{P}\left(X_n> \mathbb{E}X_n+1\right)}\right).
\end{aligned}$$

Using Chebyshev's Inequality, $\mathbb{P}\left(X_n> \mathbb{E}X_n+1\right)\le \mathrm{Var}[X_n]$. Thus

$$\mathbb{E}h(X_n)\le h(\mathbb{E}X_n)+b\mathrm{Var}[X_n]+L\cdot(\mathbb{E}X_n+1)\cdot\mathrm{Var}[X_n].$$

Because $\mathbb{E}X_n$ is bounded, $\mathbb{E}h(X_n)-h(\mathbb{E}X_n)=O(\mathrm{Var}[X_n])$. □

Proof of Theorem 4. First consider any Markov policy where $\mathrm{Var}\left[\lambda_t^u\right]=O(1/n)$ and $\mathrm{Var}\left[x_t^u\right]=O(n)$. Let $\hat{x}_t^u$ be the solution of the following differential equation:

$$\frac{d\hat{x}_t^u}{dt}=\alpha n\mathbb{E}[\lambda_t^u]\left[1+h\left(\frac{\hat{x}_t^u}{n}\right)\right]-\beta x_t^u,$$

with boundary condition $\hat{x}_0^u = 0$. We show that $\mathbb{E}[x_t^u] = \hat{x}_t^u + O(1)$. By Lagrange Theorem, we can write the variance of $h\left(x_t^u/n\right)$ as

$$\mathrm{Var}\left[h\left(\frac{x_t^u}{n}\right)\right] = \mathrm{Var}\left[h\left(\frac{x_t^u}{n}\right) - h\left(\frac{\mathbb{E}[x_t^u]}{n}\right)\right] = \mathrm{Var}\left[h'(\eta)\frac{x_t^u - \mathbb{E}[x_t^u]}{n}\right],$$

where η is a random variable between x_t^u/n and $\mathbb{E}[x_t^u]/n$. Since $h(\cdot)$ is Lipschitz continuous, $h'(\eta)$ is bounded. Thus, $\mathrm{Var}\left[h\left(\frac{x_t^u}{n}\right)\right] = O\left(\mathrm{Var}\left[\frac{x_t^u}{n}\right]\right) = O(1/n)$. Using Cauchy-Schwartz Inequality, we have:

$$\left[\mathbb{E}\lambda_t^u h\left(\frac{x_t^u}{n}\right) - \mathbb{E}\lambda_t^u \cdot \mathbb{E}h\left(\frac{x_t^u}{n}\right)\right]^2 = \mathrm{Cov}^2\left\{\lambda_t^*, h\left(\frac{x_t^u}{n}\right)\right\} \le \mathrm{Var}\left[\lambda_t^u\right] \cdot \mathrm{Var}\left[h\left(\frac{x_t^u}{n}\right)\right].$$

Since $\mathrm{Var}[\lambda_t^u] = O(1/n)$, $\mathbb{E}\lambda_t^u h\left(\frac{x_t^u}{n}\right) = \mathbb{E}\lambda_t^u \mathbb{E}h\left(\frac{x_t^u}{n}\right) + O(1/n)$. Also according to Lemma 7, $\mathbb{E}h\left(\frac{x_t^u}{n}\right) = h\left(\frac{\mathbb{E}[x_t^u]}{n}\right) + O(1/n)$. Since $\mathbb{E}[\lambda_t^u] = O(1)$, $\mathbb{E}\lambda_t^u h\left(\frac{x_t^u}{n}\right) = \mathbb{E}[\lambda_t^u] h\left(\frac{\mathbb{E}[x_t^u]}{n}\right) + O(1/n)$. Recall that from the proof of Proposition 4, $\frac{d}{dt}\mathbb{E}[x_t^u] = \alpha n \mathbb{E}[\lambda_t^u]\left[1 + h\left(\frac{x_t^u}{n}\right)\right] - \beta \mathbb{E}[x_t^u]$. Therefore,

$$\begin{aligned}
\frac{d}{dt}\left|\mathbb{E}[x_t^u] - \hat{x}_t^u\right| &= \left|\alpha n\left[\mathbb{E}\lambda_t^u h\left(\frac{x_t^u}{n}\right) - \mathbb{E}\lambda_t^u h\left(\frac{\hat{x}_t^u}{n}\right)\right] - \beta\left(\mathbb{E}[x_t^u] - \hat{x}_t^u\right)\right| \\
&\le \alpha n\left|\mathbb{E}[\lambda_t^u] h\left(\frac{\mathbb{E}[x_t^u]}{n}\right) + O(1/n) - \mathbb{E}[\lambda_t^u] h\left(\frac{\hat{x}_t^u}{n}\right)\right| + \beta\left|\mathbb{E}[x_t^u] - \hat{x}_t^u\right| \\
&\le \alpha\bar{\lambda} L\left|\mathbb{E}[x_t^u] - \hat{x}_t^u\right| + \beta\left|\mathbb{E}[x_t^u] - \hat{x}_t^u\right| + O(1).
\end{aligned}$$

The above inequality is independent of n. Thus $\mathbb{E}[x_t^u] - \hat{x}_t^u = O(1)$. Similarly,

$$\mathrm{Var}[r(\lambda_t^u)] = Var\left[r(\lambda_t^u) - r(\mathbb{E}[\lambda_t^u])\right] = \mathrm{Var}\left[r'(\xi)(\lambda_t^u - \mathbb{E}[\lambda_t^u])\right],$$

where ξ is a random variable between λ_t^u and $\mathbb{E}[\lambda_t^u]$. Since λ_t^u is bounded, $\mathrm{Var}[r(\lambda_t^u)] = O(\mathrm{Var}[\lambda_t^u]) = O(1/n)$. Thus,

$$\left[\mathbb{E}r(\lambda_t^u) h\left(\frac{x_t^u}{n}\right) - \mathbb{E}r(\lambda_t^u) \cdot \mathbb{E}h\left(\frac{x_t^u}{n}\right)\right]^2 = \mathrm{Cov}^2\left\{r(\lambda_t^u), h\left(\frac{x_t^u}{n}\right)\right\} \le \mathrm{Var}\left[r(\lambda_t^u)\right] \cdot \mathrm{Var}\left[h\left(\frac{x_t^u}{n}\right)\right].$$

Therefore $\mathbb{E}r(\lambda_t^u) h\left(\frac{x_t^u}{n}\right) = \mathbb{E}r(\lambda_t^u)\mathbb{E}h\left(\frac{x_t^u}{n}\right) + O(1/n)$. Moreover, $\mathbb{E}r(\lambda_t^u) = r(\mathbb{E}\lambda_t^u) + O(1/n)$ since λ^u is bounded. $\mathbb{E}h(x_t^u/n) = h(\mathbb{E}[x_t^u]/n) + O(1/n)$ according to Lemma 7. Thus

$$\begin{aligned}
V^u(n) &= n\mathbb{E}\int_0^T r(\lambda_t^u)\left[1 + h\left(\frac{x_t^u}{n}\right)\right] dt \\
&= n\int_0^T \left[r(\mathbb{E}[\lambda_t^u]) + \mathbb{E}h\left(\frac{x_t^u}{n}\right) r(\mathbb{E}[\lambda_t^u])\right] dt + O(1) \\
&= n\int_0^T \left[r(\mathbb{E}[\lambda_t^u]) + h\left(\frac{\hat{x}_t^u}{n}\right) r(\mathbb{E}[\lambda_t^u])\right] dt + O(1).
\end{aligned}$$

Note that $\mathbb{E}[\lambda_t^u]$ and $\hat{x}_t^u$ are a feasible solution in the deterministic problem. Therefore $V^*(n) = \max_u V^u(n) \le V^d(n) + O(1)$. On the other hand, consider the deterministic heuristic where $\lambda_t^u = \lambda_t^d$. $V^*(n) \ge V^u(n) = V^d(n) + O(1)$. Therefore, $V^*(n) = V^d(n) + O(1)$. □

Proof of Proposition 6. We prove that for any $0 \le \alpha_1^O < \alpha_2^O$ and $t < T$, $V_t(x;\alpha_1^O) < V_t(x;\alpha_2^O)$ by induction. Assume that the result holds for time $t+\delta < T$. Then $V_{t+\delta}^{\alpha_1^O}(x+\alpha_1^O) \le V_{t+\delta}^{\alpha_2^O}(x+\alpha_1^O)$, $V_{t+\delta}^{\alpha_1^O}(x-\beta\delta x+\alpha(1-\beta\delta_1)) \le V_{t+\delta}^{\alpha_2^O}(x-\beta\delta x+\alpha(1-\beta\delta_1))$, and $V_{t+\delta}^{\alpha_1^O}(x-\beta x\delta) \le V_{t+\delta}^{\alpha_2^O}(x-\beta x\delta)$. From the HJB equation,

$$\begin{aligned}
V_t(x;\alpha_1^O) = &\max_{\lambda\in[\underline{\lambda},\bar{\lambda}]}\{(h(x)+\varphi)r(\lambda)\delta + \lambda(h(x)+\varphi)\delta V_{t+\delta}(x-\beta\delta x+\alpha(1-\beta\delta_1);\alpha_1^O) \\
&+ (1-\lambda)(h(x)+\varphi)\delta V_{t+\delta}(x-\beta\delta x+\alpha_1^O(1-\beta\delta_2);\alpha_1^O) + (1-(h(x)+\varphi)\delta)[V_{t+\delta}(x-\beta\delta x;\alpha_1^O)]\} \\
\le &\max_{\lambda\in[\underline{\lambda},\bar{\lambda}]}\{(h(x)+\varphi)r(\lambda)\delta + \lambda(h(x)+\varphi)\delta V_{t+\delta}(x-\beta\delta x+\alpha(1-\beta\delta_1);\alpha_2^O) \\
&+ (1-\lambda)(h(x)+\varphi)\delta V_{t+\delta}(x-\beta\delta x+\alpha_1^O(1-\beta\delta_2);\alpha_2^O) + (1-(h(x)+\varphi)\delta)[V_{t+\delta}(x-\beta\delta x;\alpha_2^O)]\}
\end{aligned}$$

Since $V_t(x;\alpha^O)$ strictly increases in x, $V_{t+\delta}(x+\alpha_1^O;\alpha_2^O) < V_{t+\delta}(x+\alpha_2^O;\alpha_2^O)$. Thus,

$$\begin{aligned}
V_t(x;\alpha_1^O) < &\max_{\lambda\in[\underline{\lambda},\bar{\lambda}]}\{(h(x)+\varphi)r(\lambda)\delta + \lambda(h(x)+\varphi)\delta V_{t+\delta}(x-\beta\delta x+\alpha(1-\beta\delta_1);\alpha_2^O) \\
&+ (1-\lambda)(h(x)+\varphi)\delta V_{t+\delta}(x-\beta\delta x+\alpha_2^O(1-\beta\delta_2);\alpha_2^O) + (1-(h(x)+\varphi)\delta)[V_{t+\delta}(x-\beta\delta x;\alpha_2^O)]\} + o(\delta) \\
= &\, V_t(x;\alpha_2^O).
\end{aligned}$$

□

Proof of Theorem 5. We next prove part (a). Following the same step in the proof of Theorem 2, we can easily show that, if $h(x)$ is convex, then $V_t(x)$ is convex in x. Therefore, $V_t(x+\alpha)-V_t(x))$ increases in x. $\lambda_t^*(x) = \arg\max_{\lambda\in\{\lambda_1,\ldots,\lambda_K\}}\{r(\lambda)+\lambda(V_t(x+\alpha)-V_t(x))\}$. Because of the convexity of $r(\lambda)$, $\exists\, 0 \le R_{t,1} \le R_{t,2} \le \cdots \le R_{t,K-1}$, such that $\lambda_t^*(x) = \lambda_1$ if $x < R_{t,1}$; $\lambda_t^*(x) = \lambda_i$ if $R_{t,i-1} \le x < R_{t,i}$, for $i = 2,\ldots,K-1$; $\lambda_t^*(x) = \lambda_K$, otherwise.

We next prove part (b). Following the same step in the proof of corollary 1, we can easily show that, if $h(x)$ is linear and positive in $x \ge 0$, i.e., $h(x) = h_1 x + h_0$ for $h_1 \ge 0$ and $h_0 \ge 0$, then the value function $V_t(x) = \mu_t x + \omega_t$, where μ_t and ω_t satisfy the following equations

$$\frac{d\mu_t}{dt} = \beta\mu_t - h_1 \max_{\lambda\in\{\lambda_1,\ldots,\lambda_K\}}\{r(\lambda)+\lambda\alpha\mu_t\}, \tag{A.3}$$

$$\frac{d\mu_t}{dt} = \beta\mu_t + \frac{h_1}{h_0+\varphi}\frac{d\omega_t}{dt}, \tag{A.4}$$

with $\mu_T = 0$ and $\omega_T = 0$. The optimal intensity $\lambda_t^* = \arg\max_{\lambda\in\{\lambda_1,\ldots,\lambda_K\}}\{r(\lambda)+\lambda\alpha\mu_t\}$, which is independent of x.

We next prove μ_t and w_t decrease in t. By the same steps, we can prove that $V_t(x)$ decreases in t. As $V_t(x) = \mu_t x + \omega_t$, we have $\mu_t x + \omega_t$ decreases in t. By letting $x = 0$, we have ω_t decreases in t. We next prove μ_t decreases in t by contradiction. Suppose there exist $t_1 > t_2$ $\mu_{t_1} > \mu_{t_2}$, $\omega_{t_1} < \omega_{t_2}$. Let $\bar{x} = -2(\omega_{t_1}-\omega_{t_2})/(\mu_{t_1}-\mu_{t_2})$. Since $\mu_{t_1} > \mu_{t_2}$ and $\omega_{t_1} < \omega_{t_2}$, we have $\bar{x} > 0$.

Then

$$\begin{aligned} V_{t_1}(\bar{x}) - V_{t_2}(\bar{x}) &= (\mu_{t_1} - \mu_{t_2})\bar{x} + (\omega_{t_1} - \omega_{t_2}) \\ &= -(\omega_{t_1} - \omega_{t_2}) > 0, \end{aligned}$$

which contradicts proposition 2(i) that $V_t(x)$ decreases in t. Therefore μ_t decreases in t. Along with the fact that $\lambda_t^* = \arg\max_{\lambda \in \{\lambda_1, \ldots, \lambda_K\}} \{r(\lambda) + \lambda \alpha \mu_t\}$, we can have that $\lambda p'(\lambda) + p(\lambda) - c = -\alpha \mu_t$ and the left hand side decreases in λ. As μ_t decreases in t, the optimal λ_t^* increases in t. Therefore, $\exists\ 0 \le \hat{T}_1 \le \hat{T}_2 \le \cdots \le \hat{T}_{K-1} \le T$, such that $\lambda_t^*(x) = \lambda_1$ if $t < \hat{T}_1$; $\lambda_t^*(x) = \lambda_i$ if $\hat{T}_{i-1} \le t < \hat{T}_i$, for $i = 2, \ldots, K-1$; $\lambda_t^*(x) = \lambda_K$, otherwise. □

Proof of Theorem 6: We denote the optimal purchase probability under discrete prices as $\lambda_t^u(x)$ and the corresponding excitement level as x_t^u, to distinguish with the optimal solution in Problem 3. In this proof, we assume that for almost any t, the optimal purchase probability under discrete prices $\lambda_t^u(x) = \lambda_i$, for any $x \in [\mathbb{E}x_t^u - \gamma_t n, \mathbb{E}x_t^u + \gamma_t n]$, where $\gamma_t > 0$ is independent of n. This assumption ensures that $\mathbb{E}x_t^u$ is not too "close" to the boundary where the seller is indifferent between choosing purchasing probability λ_i and λ_j. We conjecture that the assumption holds for most parameters but couldn't rule out the possibility that it might be violated under special cases. Theorem 6 would still hold even without this assumption, but the analysis would be significantly more complicated as λ_t^u switches frequently between λ_i and λ_j, leading to a much larger variance of λ_t^u and x_t^u.

$$\begin{aligned} &\mathrm{Var}[\lambda_t^u] \le \mathbb{E}\left[\lambda_t^u(x_t^u) - \lambda_t^u(\mathbb{E}x_t^u)\right]^2 \\ =\ &\mathbb{E}\left[\left(\lambda_t^u(x_t^u) - \lambda_t^u(\mathbb{E}x_t^u)\right)^2 \middle| x_t^u \in [\mathbb{E}x_t^u - \gamma_t n, \mathbb{E}x_t^u + \gamma_t n]\right] \mathbb{P}\left(x_t^u \in [\mathbb{E}x_t^u - \gamma_t n, \mathbb{E}x_t^u + \gamma_t n]\right) \\ &+\mathbb{E}\left[\left(\lambda_t^u(x_t^u) - \lambda_t^u(\mathbb{E}x_t^u)\right)^2 \middle| x_t^u \notin [\mathbb{E}x_t^u - \gamma_t n, \mathbb{E}x_t^u + \gamma_t n]\right] \mathbb{P}\left(x_t^u \notin [\mathbb{E}x_t^u - \gamma_t n, \mathbb{E}x_t^u + \gamma_t n]\right) \\ \le\ &\bar{\lambda}^2 \mathbb{P}\left(x_t^u \notin [\mathbb{E}x_t^u - \gamma_t n, \mathbb{E}x_t^u + \gamma_t n]\right). \end{aligned}$$

Using Chebyshev's Inequality, $\mathrm{Var}[\lambda_t^u] = O(\mathrm{Var}[x_t^u]/n^2)$. Therefore, from Lemma 5, $\mathrm{Var}[\lambda_t^u] = O(n)$ and $\mathrm{Var}[\lambda_t^u] = O(1/n)$. $V^u(n)$ can be approximated by the profit of the deterministic problem with purchase probability $\mathbb{E}\lambda_t^u$, i.e. $V^u(n) = \Pi^d(\mathbb{E}\lambda_t^u) + O(1)$ from the proof of Theorem 4. However $\mathbb{E}\lambda_t^u = \lambda_i + O(1/n)$ for any t, whereas the optimal solution in the deterministic problem λ^d is non-stationary in t. Therefore $V^d(n) - \Pi^d(\mathbb{E}\lambda_t^u) = \Theta(n)$. Also from Theorem 4, $V^*(n) = V^d(n) + O(1))$. Thus $V^*(n) - V^u(n) = \Theta(n)$. This completes the proof. □